

QUANTITATIVE FINANCE PYTHON PROJECT

Credit Portfolio Risk Analytics

Structural Credit Risk, Rating Migration,
Dependence Modeling and Monte Carlo Simulation

A Python framework for credit portfolio simulation,
tail risk measurement and model validation

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This report documents an independent quantitative finance project. Portfolio positions, ratings, LGDs, bond terms and selected model inputs are synthetic demonstration data and do not represent actual company exposures or external credit assessments.

Abstract

This report presents a Python framework for quantitative credit portfolio risk analysis. The model combines rating and structural credit risk approaches, credit rating migration, multiple systematic factors, Gaussian and Student t copulas, and Monte Carlo simulation.

Portfolio risk is evaluated using expected loss, Value at Risk (VaR), Expected Shortfall (ES), and counterparty risk measures. The framework supports portfolios with multiple currencies, market data calibration, configurable model assumptions, and Excel portfolio inputs.

A dedicated validation framework evaluates the sensitivity of portfolio risk to dependence assumptions, Student t copula degrees of freedom, loss given default assumptions, portfolio concentration, factor structure, confidence level, and Monte Carlo sample size. Numerical stability is further assessed across different random seeds.

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1 Introduction and Scope

Credit portfolio risk is determined not only by the credit quality of individual counterparties, but also by the dependence between credit events across the portfolio. While expected credit losses are primarily driven by marginal probabilities of default and loss severity, portfolio tail risk can be strongly affected by correlated rating migrations, joint defaults, systematic risk factors, and concentration in individual counterparties [1].

This project develops a modular Python framework for the quantitative analysis of these effects. The framework combines rating and structural credit risk approaches with a credit migration model, multiple factor structures, Gaussian and Student t copulas, and Monte Carlo simulation. Portfolio risk is evaluated using expected loss, Value at Risk (VaR), Expected Shortfall (ES), and counterparty risk measures.

A central design objective is to connect theoretical credit risk concepts with a transparent implementation that can be inspected, tested, and modified. Portfolio inputs and model settings are controlled through an Excel steering workbook, while the Python library performs input validation, market data processing, probability of default estimation, factor calibration, credit migration simulation, and portfolio risk aggregation. The framework also produces structured Excel results, graphical diagnostics, and a compact PDF report.

Particular emphasis is placed on model validation and sensitivity analysis. The impact of the dependence model, Student t copula degrees of freedom, loss given default assumptions, portfolio concentration, factor structure, confidence level, and Monte Carlo sample size is examined systematically. This allows the model to be assessed in terms of numerical stability, sensitivity to assumptions, and key sources of model risk.

The framework is intended as an independent quantitative finance model development and learning project rather than a production credit risk system. Portfolio exposures, ratings, LGDs, bond characteristics, and selected model inputs are demonstration inputs and do not represent actual company positions or external credit assessments.

2 Portfolio and Input Framework

2.1 Portfolio Structure

The demonstration portfolio consists of eight counterparties across several sectors, regions, and currencies. The purpose of the portfolio is not to reproduce an actual investment portfolio, but to provide enough heterogeneity to illustrate credit migration, default risk, concentration effects, currency normalization, and systematic dependence.

The baseline portfolio contains both listed companies and synthetic counterparties. Listed firms can use observed market information for structural credit risk calibration and factor estimation, while synthetic counterparties demonstrate the use of proxy tickers and manually specified credit inputs.

Unless otherwise stated, monetary amounts in the portfolio specification and subsequent risk analysis are expressed in millions of the respective currency. For example, an exposure of 75 EUR represents EUR 75 million. After currency conversion, portfolio level monetary results are reported in USD millions.

Table 1: Baseline demonstration portfolio

Counterparty	Exposure	Currency	Sector	Region
Siemens	75	EUR	Industrial	Europe
Apple	400	USD	Technology	US
JPMorgan	90	USD	Financial	US
Roche	70	CHF	Healthcare	Europe
Shell	65	GBP	Energy	Europe
US Industrial Corp	55	USD	Industrial	US
European Tech Corp	45	EUR	Technology	Europe
Private Industrial Co	35	EUR	Industrial	Europe

The portfolio is intentionally concentrated in one large counterparty position, with Apple representing the largest exposure in the baseline configuration. This creates a useful setting for studying concentration and diversification effects. The remaining counterparties provide variation across industries, regions, currencies, credit quality assumptions, and data availability.

All exposures shown in Table 1 are expressed in their local currencies. Before portfolio aggregation, exposures are converted into a common base currency using the FX normalization framework described in Section 2.3.

2.2 Credit Risk Inputs

Each counterparty is assigned a one year probability of default (PD) using one of two credit risk approaches: a rating mapping or a structural Merton model. The selected method is controlled at counterparty level through the portfolio input workbook. This allows the framework to combine rating information with structural credit risk estimates within the same portfolio.

2.2.1 Rating Probability of Default

For counterparties using the rating approach, a credit rating is mapped to a predefined one year probability of default. The demonstration implementation uses rating categories ranging from AAA to default. These mappings are synthetic model inputs and are intended to provide a transparent credit quality hierarchy rather than reproduce the methodology or data of a specific rating agency.

The rating approach is particularly useful when a counterparty already has an available credit assessment or when a structural model based on market data is not appropriate.

2.2.2 Merton Probability of Default

For counterparties using the structural approach, the Merton model is used to estimate firm asset value and asset volatility and to derive a structural probability of default. The implementation supports three input modes:

- **Market data mode:** equity value, equity volatility, and debt inputs are retrieved from available market data.
- **Manual mode:** the required equity and debt characteristics are supplied directly through the input workbook.

- **Mixed mode:** manually supplied values are combined with market data retrieval for missing inputs.

The selected input source is recorded in the model output, improving transparency and allowing the resulting PD estimate to be traced back to its underlying calibration inputs.

Table 2: Baseline credit risk input method

Counterparty	PD Method	Credit Input
Siemens	Merton	Market data calibration
Apple	Rating	AA
JPMorgan	Rating	A
Roche	Rating	AA
Shell	Rating	BBB
US Industrial Corp	Rating	BBB
European Tech Corp	Rating	BB
Private Industrial Co	Merton	Manual calibration

This hybrid design is intentional. It allows the portfolio model to use a rating PD where a rating assessment is preferred, while retaining the ability to estimate a structural PD for counterparties without such an input or when a model estimate is desired.

Loss severity is represented through the loss given default parameter, LGD_i . In the baseline portfolio, a deterministic LGD of 50% is applied to all counterparties. The sensitivity of portfolio risk to alternative LGD assumptions is examined in Section 6.3.

2.3 Market Data and Currency Conversion

The framework combines externally retrieved market data with portfolio inputs supplied through the Excel steering workbook. External data are used for currency conversion, risk free rate proxies, market based Merton calibration and systematic factor estimation.

The portfolio supports exposures denominated in multiple currencies. In the baseline configuration, USD is used as the portfolio base currency, while individual exposures may be denominated in USD, EUR, CHF or GBP. Before portfolio level risk aggregation, each local currency exposure is converted into the selected base currency.

For counterparty i , the base currency exposure is calculated as

$$E_i^{\text{base}} = E_i^{\text{local}} \cdot FX_i, \quad (1)$$

where FX_i denotes the exchange rate from the counterparty's local currency into the portfolio base currency.

Foreign exchange rates are obtained from the euro foreign exchange reference rates published by the European Central Bank [2]. Because the ECB rates are quoted relative to the euro, the implementation derives the required cross rate when neither the local currency nor the portfolio base currency is EUR.

FX conversion is used solely to place all credit exposures on a common monetary basis. Foreign exchange risk itself is not jointly simulated with credit risk. Consequently, changes in exchange

rates during the one year simulation horizon do not contribute to the simulated portfolio loss distribution.

The framework also requires a risk free rate proxy for structural Merton calibration. The source depends on the counterparty currency. The implementation currently uses the following market inputs:

Table 3: External market data sources used by the credit portfolio model

Input	Source	Model use
FX rates	ECB reference rates	Exposure conversion
USD rate	1-year US Treasury yield	Merton calibration
EUR rate	1-year euro area AAA spot rate	Merton calibration
GBP rate	SONIA overnight rate	Merton calibration proxy
CHF rate	Swiss GMBF, 12-month	Merton calibration
Equity / ETF data	Yahoo Finance via <code>yfinance</code>	Merton and factor calibration

For USD, the model uses the one-year US Treasury constant maturity yield, series DGS1, retrieved through the Federal Reserve Economic Data database and sourced from the Board of Governors of the Federal Reserve System [3].

For EUR, the model uses the one-year spot rate from the euro area AAA government bond yield curve published through the ECB Data Portal [4].

For GBP, the implementation uses the Sterling Overnight Index Average (SONIA) as a short term risk free proxy. The series is retrieved through FRED and originates from the Bank of England [5]. SONIA is an overnight rate rather than a one-year zero coupon rate and should therefore be interpreted as a simplified proxy within the demonstration framework.

For CHF, the implementation uses an approximately twelve-month yield derived from Swiss Confederation money market debt register claim auction results published by the Federal Finance Administration [6]. The selected observation is chosen from the available auction results based on the required maturity.

Listed-company equity information and historical equity and ETF prices are accessed through the `yfinance` Python package, which retrieves publicly available Yahoo Finance market data [7, 8]. These data support both market based Merton calibration and the estimation of systematic factor loadings.

The model setting `market_data_date` controls the observation date used for FX rates and risk free rates. When the setting is `latest`, the most recent available observation is used. When a specific date is supplied, the implementation uses the latest available observation on or before the requested date. This treatment allows weekends, holidays and other non-trading days to be handled without requiring an observation on the exact requested date.

The baseline analysis uses a configured market data date of 4 September 2026.

The selected source for each FX rate and risk free rate is stored in the model output through the variables `fx_rate_source` and `risk_free_rate_source`. This provides a transparent distinction between automatically retrieved observations and fallback values.

If automatic retrieval fails, the framework can use the frozen market data snapshot stored in the input workbook. For an explicitly requested historical date, a fallback observation is used only when the fallback snapshot is not later than the requested date. This prevents a future fallback observation from being inserted into an earlier historical calculation.

The fallback snapshot is an operational robustness mechanism rather than an alternative market

data calibration. In particular, fallback risk free rates are simplified flat rate proxies and do not represent complete zero coupon yield curves.

The `market_data_date` setting should also not be interpreted as a complete point in time valuation date for every external model input. Equity information used for market based Merton calibration and historical price series used for factor estimation retain their respective rolling data retrieval procedures. Consequently, the framework improves reproducibility of FX and risk free inputs but does not provide a fully historical reconstruction of all market information.

These market data simplifications, together with the absence of simulated FX risk and complete interest rate term structures, are discussed further in Section 7.

2.4 Factor Proxies

Systematic dependence between counterparties is represented through observable market factor proxies. Historical equity and ETF price data are retrieved from Yahoo Finance through the `yfinance` Python package [7, 8].

The baseline factor structure combines global, regional and sector components. The following exchange traded funds are used as market proxies:

Table 4: Systematic factor proxies

Factor	Ticker	Proxy
Global	ACWI	MSCI All Country World
United States	SPY	S&P 500
Europe	VGK	European equities
Technology	IXN	Global technology equities
Industrial	EXI	Global industrial equities
Financial	IXG	Global financial equities
Energy	IXC	Global energy equities
Healthcare	IXJ	Global healthcare equities

Factor loadings are estimated from historical equity returns over the configured lookback period. For each listed counterparty, the company’s own equity return series is used where available. The relationship between the counterparty return and the selected systematic factor returns provides the basis for estimating its exposure to global, regional and sector risk.

For counterparties without a directly observable listed equity series, the framework allows a listed proxy ticker to be supplied through the input workbook. This makes it possible to include synthetic or private counterparties while retaining an empirically estimated dependence structure. The selected proxy is recorded in the model input and should be interpreted as an approximation rather than as an exact representation of the counterparty’s systematic risk.

The factor structure is configurable. The implementation supports a single global factor, global and sector factors, global and region factors, and the complete global, sector and region specification used in the baseline model. The sensitivity of portfolio VaR and Expected Shortfall to these alternative structures is examined in Section 6.5.

The selected factor proxies and estimated loadings provide a transparent mechanism for introducing economically interpretable systematic dependence into the portfolio model. They should nevertheless be regarded as modeling choices. Equity market comovement is only a proxy for dependence in credit migration and default, and alternative indices, lookback periods or proxy

companies could lead to different estimated relationships. These limitations are discussed further in Section 7.

3 Credit Risk Methodology

3.1 Expected Loss

For counterparty i , the one year expected default loss is defined as

$$EL_i = E_i \cdot PD_i \cdot LGD_i, \quad (2)$$

where E_i denotes the exposure of counterparty i , PD_i its probability of default over the relevant horizon, and LGD_i the corresponding loss given default.

The portfolio expected default loss is obtained by aggregation across all counterparties:

$$EL_{\text{portfolio}} = \sum_{i=1}^n E_i \cdot PD_i \cdot LGD_i. \quad (3)$$

Expected default loss represents the average loss attributable to default risk under the marginal counterparty PD assumptions. It is therefore a quantity based only on exposure, probability of default, and loss severity.

This quantity is distinct from the mean loss generated by the portfolio migration simulation. The simulated portfolio loss distribution also captures changes in market value caused by rating upgrades and downgrades, as well as losses resulting from default. Consequently, the mean simulated portfolio loss need not coincide with the portfolio expected default loss defined above.

This distinction is important throughout the analysis: the portfolio model includes both rating migration and default rather than modeling default alone.

The expected default loss defined above should also be distinguished from the unexpected loss measure used later in the portfolio risk analysis. In this implementation, unexpected loss is defined relative to the simulated migration loss distribution as the difference between portfolio Value at Risk and the mean simulated portfolio loss. The formal definition is introduced together with the portfolio risk measures in Section 3.7.

3.2 Structural Credit Risk: Merton Model

The structural credit risk component of the framework is based on the structural corporate debt model introduced by Merton [9]. The central idea is to interpret the market value of a firm's equity as a European call option on the value of the firm's assets. Shareholders have limited liability and therefore receive a positive payoff only when the value of the firm's assets exceeds its debt obligation at maturity.

Let

- V denote the current market value of firm assets,
- E the current market value of equity,
- D the debt obligation or default barrier,

- σ_V the volatility of firm assets,
- σ_E the volatility of equity,
- r the risk free interest rate, and
- T the model horizon.

At maturity, equity has the payoff

$$E_T = \max(V_T - D, 0). \quad (4)$$

This is identical to the payoff of a European call option with the firm's asset value as the underlying and the debt obligation as the strike price. Default occurs when the firm's asset value is insufficient to meet the debt obligation,

$$V_T \leq D. \quad (5)$$

Figure 1 illustrates this relationship.

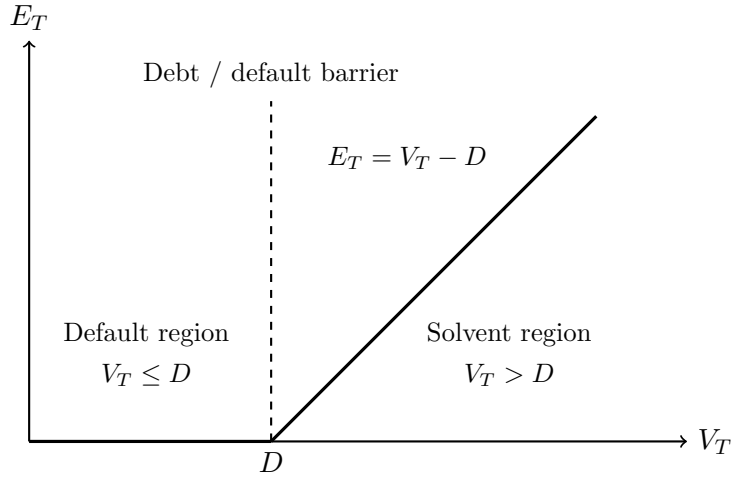


Figure 1: Equity payoff in the Merton structural credit risk model. Equity behaves like a call option on firm assets, while default occurs when the asset value at maturity does not exceed the debt obligation.

3.2.1 Equity Valuation

Under the standard Merton assumptions, the current value of equity can be written using the Black-Scholes option pricing formula [10, 9]:

$$E = VN(d_1) - De^{-rT}N(d_2), \quad (6)$$

where

$$d_1 = \frac{\ln(V/D) + (r + \frac{1}{2}\sigma_V^2)T}{\sigma_V\sqrt{T}}, \quad (7)$$

and

$$d_2 = d_1 - \sigma_V \sqrt{T}. \quad (8)$$

Here, $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

3.2.2 Equity and Asset Volatility

Firm asset value and asset volatility are not directly observable. Equity volatility therefore provides a second relationship that can be used to infer the latent asset parameters.

Using the Black–Scholes equity delta, the relationship between equity volatility and asset volatility is

$$\sigma_E E = N(d_1) \sigma_V V. \quad (9)$$

The model therefore provides two nonlinear equations,

$$E = V N(d_1) - D e^{-rT} N(d_2), \quad (10)$$

$$\sigma_E E = N(d_1) \sigma_V V, \quad (11)$$

with the two unknown quantities V and σ_V .

The implementation solves this nonlinear system numerically to recover the latent firm asset value and asset volatility from observable or manually specified equity and debt information.

3.2.3 Structural Probability of Default

Once V and σ_V have been calibrated, the structural probability of default used in the implementation is calculated as

$$PD_{\text{Merton}} = N(-d_2). \quad (12)$$

Because the implemented d_2 is constructed using the risk free rate, this quantity should be interpreted as a model implied structural default probability under the valuation assumptions of the Merton framework rather than as an empirically calibrated physical default probability. The quantity d_2 captures the standardized distance between the firm’s asset value and its debt boundary under the assumptions of the Merton framework. A larger value of d_2 therefore corresponds to a lower probability that firm assets fall below the debt obligation at the selected horizon.

This structural interpretation differs fundamentally from the rating approach. Rather than assigning a PD from a discrete credit category, the Merton model derives credit risk from the relationship between market equity value, equity volatility, leverage, interest rates, and the model horizon.

3.2.4 Calibration Modes

The implementation supports three alternative sources for the inputs required by the structural model:

- **Market data calibration:** equity value, equity volatility, and debt information are obtained from available market data.
- **Manual calibration:** the required equity value, equity volatility, and debt inputs are supplied directly through the portfolio input workbook.
- **Mixed calibration:** manually supplied values are retained while missing inputs are completed using available market data.

This design allows the same structural methodology to be used for both listed companies and demonstration counterparties for which market inputs may be unavailable or intentionally supplied manually. The calibration source is retained in the model output to make the resulting PD estimate traceable.

3.2.5 Interpretation and Limitations

The Merton model provides a transparent economic link between capital structure, market volatility, and credit risk. It is particularly useful because the latent firm asset process can be inferred from observable equity information rather than requiring a predefined external rating.

However, structural model outputs must be interpreted carefully. For firms with high market capitalization relative to debt and moderate equity volatility, the calibrated distance to default can become very large, producing extremely small structural PDs. Such values reflect the assumptions of the model and should not automatically be interpreted as realistic long run default probabilities or external credit assessments.

This issue appears directly in the demonstration portfolio, where some Merton estimates based on market data can be substantially lower than rating default probabilities. These results should therefore be interpreted in the context of the basic Merton specification implemented in this project. More advanced structural credit risk models extend the original framework by relaxing assumptions on the default mechanism, capital structure, interest rates, or asset dynamics. The structural PD produced here should therefore be viewed as one model based credit signal rather than as an unquestioned measure of true counterparty creditworthiness.

The model also assumes a simplified capital structure and a single debt barrier at the selected horizon. More realistic applications may require richer debt structures, alternative default barriers, stochastic interest rates, jump risk, or other extensions. These limitations are revisited in the dedicated model limitations section.

3.3 Rating Credit Risk

For counterparties using the rating approach, the model assigns a one year probability of default according to the selected credit rating. The implementation uses a synthetic mapping between rating categories and default probabilities.

Conceptually, the mapping can be written as

$$PD_i = f(R_i), \tag{13}$$

where R_i denotes the credit rating of counterparty i and $f(\cdot)$ is the rating to PD mapping used by the model.

In the demonstration portfolio, the mapping is intentionally simplified and does not reproduce the methodology of a specific external rating agency. The purpose is to provide a transparent hierarchy of credit quality that can be combined with the structural Merton approach within the same portfolio.

The rating framework is particularly useful for counterparties for which a discrete credit assessment is already available or where structural calibration using market data is not appropriate. It also provides the starting point for the credit migration mechanism described in the following section.

For counterparties using the Merton model, the structural PD is translated into an equivalent rating category for the purpose of migration simulation. This creates a common migration framework across rating and structurally calibrated counterparties. The mapping is an approximation and therefore represents an important source of model risk.

3.4 Credit Rating Migration

The portfolio model captures not only default events but also changes in credit quality over the simulation horizon. Credit migration is represented using a one year rating transition matrix and a latent variable threshold construction similar to the CreditMetrics framework [11].

For each initial rating category r , the transition matrix specifies the probabilities

$$p_{r,j} = \mathbb{P}(R_{t+1} = j \mid R_t = r), \quad (14)$$

where j denotes a possible rating state at the end of the horizon, including default.

For every initial rating, the transition probabilities satisfy

$$\sum_j p_{r,j} = 1. \quad (15)$$

The transition matrix used in the demonstration framework is synthetic and is intended to reproduce plausible migration behavior rather than replicate the historical transition data of a specific rating agency.

3.4.1 Simulation of Rating States

For each counterparty and Monte Carlo scenario, a latent credit variable is simulated. The simulated value is mapped to a rating outcome using thresholds derived from the cumulative transition probabilities of the counterparty's initial rating.

For an initial rating r , define cumulative probabilities

$$C_{r,k} = \sum_{j \leq k} p_{r,j}. \quad (16)$$

These cumulative probabilities are transformed into thresholds for the latent variable. This threshold construction follows the standard idea of mapping transition probabilities to regions of a latent credit variable [11]. The simulated systematic and idiosyncratic credit factors then determine which rating state is reached in each scenario.

This construction allows systematic dependence between counterparties to influence rating migrations while preserving the marginal transition probabilities implied by the transition matrix.

3.4.2 Migration Losses

A rating migration does not necessarily imply default. Rating upgrades and downgrades can also change the market value of a credit exposure. The use of migration state dependent bond revaluation follows the general CreditMetrics approach, in which changes in rating are associated with changes in credit spreads and therefore changes in the present value of remaining cash flows [11].

For migrations that do not result in default, the framework approximates the change in value through rating dependent credit spreads. A downgrade generally increases the relevant credit spread and reduces the present value of the position, while an upgrade can produce a gain in market value.

If counterparty i migrates from rating r to rating j , the migration loss can be represented conceptually as

$$L_i^{\text{migration}} = V_i(r) - V_i(j), \quad (17)$$

where $V_i(r)$ is the value of the position under its initial rating and $V_i(j)$ the value under the simulated rating at the end of the horizon.

For a default state, the loss is determined by the exposure and loss given default assumption,

$$L_i^{\text{default}} = E_i \cdot LGD_i. \quad (18)$$

The simulated counterparty loss therefore combines migration and default effects, and the portfolio loss in scenario s is obtained by aggregation:

$$L^{(s)} = \sum_{i=1}^n L_i^{(s)}. \quad (19)$$

This representation explains why counterparties can contribute materially to portfolio VaR and Expected Shortfall even when their one year default probability is very small. In particular, a counterparty with a low structural PD can still generate losses through adverse rating migration and spread widening.

The final simulated loss distribution therefore reflects both default severity and changes in credit quality, making it richer than a model based only on default events.

3.5 Systematic Factor Model

Using the factor return series and proxy selection described in Section 2.4, the model estimates how strongly each counterparty is exposed to common systematic risk factors.

The factor sensitivities are estimated from historical equity returns. Conceptually, the estimation can be represented by

$$r_{i,t} = \alpha_i + \beta_{i,GR}GR_{G,t} + \beta_{i,RR}RR_{R,t} + \beta_{i,SR}SR_{S,t} + u_{i,t}, \quad (20)$$

where $r_{i,t}$ denotes the return of counterparty i or its selected proxy, $r_{G,t}$ the global factor return, $r_{R,t}$ the relevant regional factor return, $r_{S,t}$ the relevant sector factor return, and $u_{i,t}$ the remaining idiosyncratic component.

The estimated coefficients measure the sensitivity of the counterparty to the corresponding systematic factors. Only the factors included in the selected model structure enter the estimation.

For the credit simulation, the systematic and idiosyncratic components are combined into a latent credit variable,

$$X_i = \beta_{i,G}G + \beta_{i,R}R_i + \beta_{i,S}S_i + \beta_{i,\varepsilon}\varepsilon_i, \quad (21)$$

where G , R_i , and S_i denote the simulated global, regional, and sector factors, while ε_i represents counterparty specific risk.

The latent variable X_i is then mapped to the rating migration thresholds introduced in Section 3.4. Shared systematic factors therefore create dependence between the migration outcomes of different counterparties: firms with similar factor exposures can experience adverse credit movements simultaneously when common factors deteriorate.

The framework supports four factor structures:

- **Single factor:** only the global factor is included;
- **Global + sector:** global and sector factors are included;
- **Global + region:** global and regional factors are included;
- **Global + sector + region:** global, sector, and regional factors are included.

The global, sector, and region specification is used as the baseline model. The alternative specifications are compared later in the validation analysis to assess the sensitivity of portfolio VaR and Expected Shortfall to the assumed factor structure.

3.6 Dependence Modeling

The systematic factor model described in Section 3.5 determines which common risk factors affect each counterparty. A separate dependence assumption determines how the resulting credit variables behave jointly across the portfolio.

The framework considers three dependence specifications: independent credit outcomes, a Gaussian copula, and a Student t copula. The marginal rating transition probabilities of each counterparty remain unchanged across these specifications. What changes is the dependence between counterparties and, in particular, the probability that several adverse credit movements occur at the same time.

This separation between marginal behavior and dependence is a central feature of copula modeling [1]. It is particularly important for credit portfolio risk because similar individual credit probabilities can produce very different portfolio tail losses under different assumptions about joint adverse events.

3.6.1 Independent Credit Events

The independent specification provides the simplest benchmark. Credit variables are generated independently across counterparties, so the simulated credit outcome of one counterparty does

not affect the outcome of another.

Conceptually,

$$U_i \sim U(0, 1), \quad (22)$$

with

$$U_i \perp U_j \quad \text{for } i \neq j. \quad (23)$$

Each simulated value is mapped to the rating migration thresholds of the corresponding counterparty. The marginal transition probabilities are therefore preserved, but there is no dependence between the simulated credit outcomes.

This specification serves as a useful benchmark. Comparing it with the Gaussian and Student t specifications shows how much additional portfolio tail risk is introduced by dependence between counterparties.

3.6.2 Gaussian Copula

The Gaussian copula introduces dependence through a multivariate normal structure [1]. Let

$$Z = (Z_1, \dots, Z_n)^\top \sim N(0, R), \quad (24)$$

where R represents the correlation structure generated by the systematic factor model.

The latent normal variables are transformed into uniform variables using the standard normal cumulative distribution function,

$$U_i = \Phi(Z_i). \quad (25)$$

The resulting uniform variables are then mapped to the rating migration thresholds of the individual counterparties.

This construction separates the marginal credit behavior from the dependence structure. The transition matrix determines the individual rating probabilities, while the Gaussian copula determines how the credit outcomes occur jointly across the portfolio.

The Gaussian copula therefore allows correlated rating migrations and defaults. However, for correlations below one, it has zero asymptotic tail dependence [1]. This does not mean that extreme observations cannot occur together. Rather, the dependence between increasingly extreme observations weakens in the far tail.

For credit portfolio risk, this can be important because the most severe portfolio losses are often associated with several counterparties deteriorating at the same time.

3.6.3 Student t Copula

The Student t copula provides an alternative dependence structure that allows stronger dependence during extreme scenarios [12]. It uses the same underlying correlation structure R but introduces an additional parameter, the degrees of freedom ν .

A useful representation of the multivariate Student t distribution is

$$T = \frac{Z}{\sqrt{W/\nu}}, \quad (26)$$

where

$$Z \sim N(0, R), \quad W \sim \chi_\nu^2, \quad (27)$$

and Z and W are independent [12].

The important feature is that all components of Z are divided by the same random quantity,

$$\sqrt{W/\nu}. \quad (28)$$

This creates a common scaling shock across all counterparties. If the denominator becomes unusually small, several credit variables are magnified at the same time. The Student t structure can therefore generate joint extreme outcomes more easily than the Gaussian structure.

3.6.4 Role of the Degrees of Freedom

The degrees of freedom ν control how strongly the common scaling factor can vary.

The scaled variable W/ν has

$$\mathbb{E} \left[\frac{W}{\nu} \right] = 1 \quad (29)$$

and

$$\text{Var} \left(\frac{W}{\nu} \right) = \frac{2}{\nu}. \quad (30)$$

The expected scale is therefore the same for different values of ν , but its variability changes substantially.

For a small value of ν , the random scale can move far away from one. In some scenarios, W/ν becomes small, which makes the denominator $\sqrt{W/\nu}$ small and magnifies all components of T simultaneously. This produces heavier joint tails and increases the probability of several counterparties experiencing extreme credit movements together.

For a larger value of ν , the random scale is more concentrated around one. The common amplification effect becomes weaker and the dependence structure becomes closer to the Gaussian case.

This gives the degrees of freedom a direct interpretation in the portfolio model:

- **Low ν :** more variable common scaling, heavier joint tails, and stronger extreme dependence;
- **High ν :** more stable common scaling and behavior closer to Gaussian dependence.

The Student t variables are transformed into uniform variables using the Student t cumulative distribution function,

$$U_i = F_\nu(T_i), \quad (31)$$

where F_ν denotes the cumulative distribution function of a Student t distribution with ν degrees of freedom.

These uniform variables are then mapped to the same marginal rating transition probabilities as under the other dependence specifications. Changing ν therefore changes how counterparties behave jointly without changing their individual migration probabilities.

3.6.5 Tail Dependence

Unlike the Gaussian copula, the Student t copula exhibits nonzero tail dependence [12]. Tail dependence measures the tendency of variables to become extreme together.

For two variables with correlation ρ and ν degrees of freedom, the upper and lower tail dependence coefficients of the symmetric Student t copula are

$$\lambda_U = \lambda_L = 2F_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right), \quad (32)$$

where $F_{\nu+1}$ denotes the cumulative distribution function of a Student t distribution with $\nu+1$ degrees of freedom.

The expression shows that tail dependence depends on both the correlation ρ and the degrees of freedom ν . For a given correlation, lower degrees of freedom produce stronger tail dependence.

The baseline model uses a Student t copula with five degrees of freedom. Values of $\nu = 3$, $\nu = 5$, and $\nu = 10$ are compared later in the validation analysis. This directly tests how different strengths of joint tail behavior affect portfolio VaR and Expected Shortfall.

3.7 Portfolio Risk Measures

The simulated portfolio loss distribution is summarized using Value at Risk (VaR), Expected Shortfall (ES), and Unexpected Loss (UL). VaR and Expected Shortfall are standard tail risk measures; the definitions used here follow McNeil, Frey and Embrechts [1].

3.7.1 Value at Risk

Value at Risk represents the loss threshold that is exceeded only with probability $1 - \alpha$. At confidence level α , it is defined as

$$VaR_\alpha(L) = \inf \{l : \mathbb{P}(L \leq l) \geq \alpha\}. \quad (33)$$

For example, a 99.5% VaR represents a loss level that is exceeded in approximately 0.5% of simulated scenarios.

VaR provides information about the location of the loss distribution at the selected quantile, but it does not describe the severity of losses beyond that threshold.

3.7.2 Expected Shortfall

Expected Shortfall measures the average loss in the extreme tail beyond the selected confidence level. A general representation in terms of the loss quantile function is

$$ES_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 VaR_u(L) du. \quad (34)$$

For a continuous loss distribution with no probability mass at the VaR threshold, this reduces to the conditional expectation

$$ES_\alpha(L) = \mathbb{E}[L \mid L \geq VaR_\alpha(L)]. \quad (35)$$

The second representation provides the intuitive interpretation of Expected Shortfall: it is the average loss conditional on being in the worst $1 - \alpha$ fraction of the loss distribution.

In the Monte Carlo implementation, Expected Shortfall is estimated directly from the simulated loss distribution by averaging the worst $1 - \alpha$ fraction of portfolio loss scenarios.

At a confidence level of 99.5%, for example, ES therefore represents the average loss among approximately the worst 0.5% of simulated outcomes.

Expected Shortfall complements VaR by describing the severity of losses beyond the selected quantile rather than only identifying the loss threshold itself.

3.7.3 Unexpected Loss

Unexpected Loss measures the difference between the selected VaR and the average simulated portfolio loss. In this implementation it is defined as

$$UL_\alpha = VaR_\alpha(L) - \mathbb{E}[L]. \quad (36)$$

Here, $\mathbb{E}[L]$ refers to the mean of the simulated migration and default loss distribution. It should therefore not be confused with the expected default loss defined in Section 3.1, which is calculated directly from exposure, probability of default, and loss given default.

Unexpected Loss represents the additional loss above the average simulated outcome at the selected confidence level.

3.8 Counterparty Risk Contributions

Portfolio VaR and Expected Shortfall describe the risk of the portfolio as a whole, but they do not identify which counterparties are responsible for that risk. The framework therefore calculates several counterparty risk measures based on the same simulated loss scenarios used for the portfolio calculation.

Let

$$L = \sum_{j=1}^n L_j \quad (37)$$

denote the total portfolio loss, and let

$$L_{-i} = L - L_i \quad (38)$$

denote the portfolio loss after removing counterparty i .

The notation $\rho(\cdot)$ is used below for the selected portfolio risk measure, which can be either VaR or Expected Shortfall at the chosen confidence level.

3.8.1 Standalone Risk

The standalone risk of counterparty i is the risk of its simulated loss distribution when considered independently of the remaining portfolio:

$$\rho_i^{\text{standalone}} = \rho(L_i). \quad (39)$$

Standalone risk therefore measures the tail risk of the counterparty itself without taking diversification with the rest of the portfolio into account.

3.8.2 Incremental Risk

Incremental risk measures how much total portfolio risk changes when counterparty i is removed:

$$\rho_i^{\text{incremental}} = \rho(L) - \rho(L_{-i}). \quad (40)$$

A large positive incremental risk indicates that the counterparty has a substantial effect on portfolio tail risk. The measure reflects not only the counterparty's own credit risk but also its dependence with the remaining portfolio.

This makes incremental risk particularly useful for identifying concentrations that may not be apparent from exposure size or standalone risk alone.

3.8.3 Marginal Risk

In this implementation, marginal risk is defined as incremental risk normalized by the counterparty exposure:

$$\rho_i^{\text{marginal}} = \frac{\rho_i^{\text{incremental}}}{E_i}. \quad (41)$$

The measure therefore expresses the change in portfolio risk per unit of counterparty exposure. It allows counterparties of different sizes to be compared on a common exposure basis.

This makes the measure particularly useful for portfolio steering and capital efficiency analysis. A counterparty with a high marginal risk measure consumes relatively more portfolio tail risk per unit of exposure than a counterparty with a lower value. The metric can therefore help identify exposures that are inefficient from a risk perspective and support comparisons between alternative portfolio positions.

The term marginal risk should be interpreted according to this implementation. It is a leave one out measure divided by exposure and is not an Euler derivative or Euler risk contribution. It should therefore be viewed as a comparative risk diagnostic rather than as an exact allocation of portfolio capital.

3.8.4 Diversification Benefit

The diversification benefit of counterparty i is defined as the difference between its standalone risk and its incremental contribution to portfolio risk:

$$DB_i = \rho_i^{\text{standalone}} - \rho_i^{\text{incremental}}. \quad (42)$$

A positive diversification benefit indicates that the counterparty contributes less risk within the portfolio than it would generate on a standalone basis. This can arise because its losses are not perfectly aligned with losses from the other counterparties.

Conversely, a small or negative diversification benefit can indicate that the counterparty is strongly exposed to the same adverse scenarios as the rest of the portfolio.

3.8.5 Interpretation

These measures provide different views of counterparty risk. Standalone risk describes the counterparty in isolation, incremental risk measures its effect on total portfolio risk, and marginal risk expresses that effect relative to its exposure.

The measures are calculated for both VaR and Expected Shortfall. In particular, incremental ES can be useful for identifying counterparties that contribute strongly to severe portfolio loss scenarios even when their probability of default is relatively small.

The incremental measures used here should not be interpreted as an additive allocation of portfolio risk. Because VaR and Expected Shortfall are nonlinear functions of the portfolio loss distribution, the leave one out incremental risks of the individual counterparties do not generally sum to total portfolio VaR or Expected Shortfall.

The same applies to the exposure normalized marginal measures. Their purpose is counterparty comparison and risk diagnostics rather than exact allocation of total portfolio risk.

4 Implementation Architecture

The project is designed as a modular Python library with a clear separation between reusable quantitative models, executable workflows, input data, testing, and generated results. This separation allows individual model components to be tested independently while keeping portfolio configuration and reporting outside the core library.

The credit portfolio framework follows the same architecture as the option pricing and numerical methods developed earlier in the project. Reusable financial models are kept inside the `quant_finance` package, while executable portfolio workflows and report generation are handled separately.

4.1 Repository Structure

The final project is organized around the following top level structure:

```
quant-finance-library/  
|  
+-- quant_finance/
```

```

|   +-- options/
|   +-- credit/
|
+-- examples/
+-- tests/
+-- data/
+-- outputs/
+-- docs/
+-- README.md

```

The purpose of each component is summarized in Table 5.

Table 5: Repository structure and responsibilities

Component	Purpose
<code>quant_finance/</code>	Reusable Python package containing the quantitative models and analytical functions. Core model logic is kept independent of reporting, plotting, and demonstration inputs.
<code>examples/</code>	Executable workflows that combine the library components. This includes the portfolio model run, validation workflow, report generation, and graphical output.
<code>tests/</code>	Automated tests implemented with <code>pytest</code> . The tests verify individual functions, numerical results, input validation, simulation behavior, and interactions between model components.
<code>data/</code>	Input files used to steer the portfolio model, including the Excel portfolio workbook and its model configuration.
<code>outputs/</code>	Generated Excel results, PDF reports, plots, scenario losses, and validation experiment outputs.
<code>docs/</code>	Technical documentation for the project, including the final quantitative model report.

Within the reusable package, the `options` and `credit` modules remain separate. The former contains the option pricing, simulation, and hedging components developed in the earlier stages of the project, while the latter contains the credit risk functionality used by the portfolio model.

An important design principle is that core library modules do not contain hard coded demonstration portfolios, plots, or reporting logic. This keeps the quantitative functions reusable and allows them to be tested independently.

4.2 Portfolio Input and Configuration

The credit portfolio model is controlled through an Excel steering workbook rather than through hard coded portfolio assumptions in the Python source code. The workbook contains five main input areas:

- `credit_portfolio`: counterparty exposures, currencies, ratings, Merton inputs, bond characteristics, and model choices;

- **model_settings**: simulation settings, confidence level, dependence model, Student t degrees of freedom, factor structure, base currency, random seed, and output configuration;
- **factor_proxies**: global, regional, and sector market proxies used for factor estimation;
- **rating_migration_matrix**: the transition probabilities used for rating migration simulation;
- **market_data**: market inputs and fallback values required by the portfolio model.

Separating these inputs from the Python implementation makes the portfolio model easier to modify and inspect. Portfolio composition and important model assumptions can be changed without editing the core quantitative code.

Before a simulation is performed, the inputs are validated for required fields, supported model choices, numerical ranges, and internal consistency. This prevents invalid portfolio or model configurations from silently propagating into the simulation.

4.3 Model Execution Workflow

The main portfolio workflow connects the individual model components in a defined sequence. The following figure provides an overview of the complete execution process.

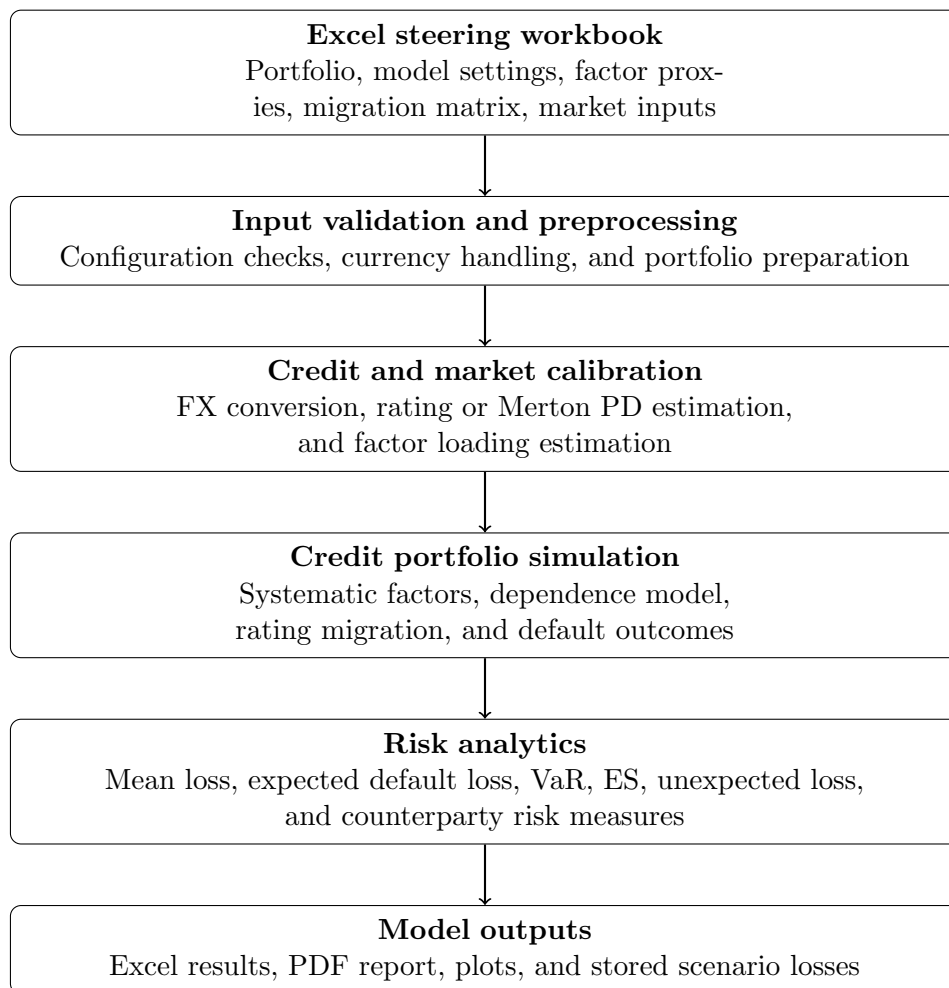


Figure 2: Execution workflow of the credit portfolio model.

The workflow begins with portfolio and model configuration rather than with assumptions embedded in the code. After validation, exposures are converted into the selected base currency and the required credit inputs are determined. Depending on the counterparty, the probability of default is obtained from either the rating framework or the Merton model.

Historical market data are then used to estimate the systematic factor sensitivities required by the selected factor structure. The dependence model and factor structure are combined with the rating transition mechanism to generate credit outcomes for each Monte Carlo scenario.

Counterparty losses are aggregated into a portfolio loss distribution from which the portfolio risk measures described in Section 3.7 are calculated. The same scenarios are also used to calculate the counterparty risk measures described in Section 3.8.

4.4 Core Library and Reporting Separation

The normal portfolio workflow is executed through the scripts in the `examples` directory, while the underlying calculations remain inside the `quant_finance` package.

For example, the main credit portfolio workflow can be executed through

```
python examples/credit_portfolio_output.py
```

The executable script coordinates input loading, model execution, and output creation. It does not replace the underlying quantitative library. Portfolio simulation, credit analytics, calibration, and other reusable calculations remain separate from the reporting layer.

The reporting workflow creates structured Excel outputs, graphical diagnostics, and a compact PDF summary. A separate reporting module is responsible for the PDF generation, while the quantitative functions themselves remain independent of the presentation format.

This separation has two advantages. First, the same quantitative functions can be reused in other applications without the existing Excel or PDF reporting layer. Second, the numerical model can be tested independently from presentation and file generation.

4.5 Testing and Model Validation

Automated software testing and quantitative model validation are treated as separate but complementary parts of the project.

Automated tests verify the behavior of individual functions and model components. These include tests for pricing functions, numerical methods, credit rating logic, portfolio inputs, simulation behavior, risk measures, and important numerical edge cases. The full test suite can be executed using

```
pytest
```

The credit portfolio tests also cover different dependence assumptions and factor structures rather than testing only the baseline configuration.

Model validation addresses a different question. Instead of checking whether the code performs a calculation as intended, it examines how the resulting portfolio risk changes when important model assumptions are varied.

A dedicated validation workflow therefore evaluates dependence assumptions, Student t degrees of freedom, LGD assumptions, portfolio concentration, factor structure, confidence level, Monte Carlo sample size, and random seed stability.

The validation experiments are kept separate from the baseline portfolio workflow. Their results are aggregated through

```
python examples/credit_portfolio_validation.py
```

This distinction between software testing and model validation is intentional. Passing unit tests provides evidence that the implementation behaves consistently with its specification, while sensitivity analysis and numerical stability checks provide evidence about the behavior and limitations of the model itself.

4.6 Reproducibility and Auditability

Reproducibility and auditability are supported through centralized model settings, explicit random seeds, retained input data, source indicators, deterministic test cases and automated validation.

The Monte Carlo simulation uses a user defined random seed. As long as the model inputs, simulation settings and implementation remain unchanged, the same random scenarios can therefore be reproduced. This allows numerical results to be compared consistently across model changes, validation exercises and sensitivity analyses.

The Excel steering workbook provides a central record of the main model assumptions and configuration choices. These include the number of simulations, confidence level, dependence model, degrees of freedom for the t -copula, factor structure, factor lookback period, base currency and output settings. Keeping these assumptions in a single steering file reduces the need for hard coded parameters inside the model implementation.

For automatically retrieved foreign exchange rates and risk free rates, the framework also includes the setting

`market_data_date.`

The value `latest` instructs the model to use the latest available observation. Alternatively, a specific date may be entered in the form `YYYY-MM-DD`. In that case, the model selects the latest available market observation on or before the requested date. This allows the model to handle weekends, public holidays and other dates for which no exact market observation exists.

The requested market data date is retained in the model output so that the intended market data reference can be identified for each run. However, the setting currently applies specifically to FX rates and risk free rates. It is therefore not intended to represent a complete historical valuation date for all model inputs. Equity market inputs used in the Merton model and historical data used for factor loading estimation retain their own data retrieval procedures and rolling historical windows.

Automatic market data retrieval and fallback data are kept conceptually separate. Automatic retrieval is attempted first. If it fails, the framework may use the frozen backup values stored in the Excel steering workbook.

The fallback snapshot contains its own `as_of_date`. This date refers only to the stored backup data and is distinct from the requested `market_data_date`. In the generated Excel output, the

backup values are reported separately in the `fallback_market_data` sheet in order to make this distinction explicit.

For historical market data requests, the fallback mechanism includes an additional consistency check. A fallback snapshot with an `as_of_date` later than the requested `market_data_date` is rejected. This prevents future market information from entering a run that is intended to represent an earlier date.

The source used for individual market inputs is retained at counterparty level. The fields

`fx_rate_source` and `risk_free_rate_source`

identify whether the model used automatically retrieved market data, the frozen fallback snapshot or, for FX conversion, the portfolio base currency treatment. This provides a direct audit trail of the market data path used in the calculation.

The model output also retains intermediate and counterparty level information such as probabilities of default, PD sources, converted exposures, factor calibration tickers, factor sources, calibrated factor loadings, expected losses and counterparty risk contributions. This allows important stages of the calculation to be inspected rather than presenting only final portfolio risk measures.

Automated tests provide an additional reproducibility and control layer. Unit tests cover the individual pricing, credit risk, migration, dependence, market data and portfolio components. Dedicated tests also verify the market data date logic, including historical date propagation, previous available observation selection and protection against using fallback information from after the requested date.

Model validation is kept separate from the baseline model execution. The dedicated validation workflow evaluates alternative dependence assumptions, t -copula degrees of freedom, LGD assumptions, concentration effects, factor structures, confidence levels and Monte Carlo stability. This separation allows the baseline model to remain compact while retaining a structured framework for challenging the main assumptions and numerical behavior.

Taken together, the centralized steering file, deterministic simulation seed, explicit market data source tracking, frozen fallback snapshot, automated tests and separate validation workflow provide a transparent record of the principal assumptions, data sources and calculation paths used in each model run. At the same time, the framework does not claim full point in time reproducibility for all external market inputs, since some equity and factor data continue to rely on their own rolling market data procedures.

5 Baseline Portfolio Results

This section presents the results of the baseline credit portfolio configuration. The purpose is to illustrate how the different model components interact in a complete portfolio run and to establish a reference case for the validation and sensitivity analysis in Section 6.

The baseline results should be interpreted as a demonstration of the framework rather than as estimates for an actual investment portfolio. Counterparty exposures, ratings, LGDs, bond characteristics and several other portfolio inputs are synthetic or selected for illustrative purposes.

5.1 Baseline Model Configuration

The baseline portfolio contains eight counterparties across several industries, regions and currencies. The portfolio combines rating based credit risk inputs with structural Merton based probabilities of default and includes both listed and synthetic private counterparties.

The principal model settings used for the baseline calculation are summarized in Table 6.

Table 6: Baseline model configuration

Setting	Baseline value
Number of simulations	100,000
Random seed	0
Confidence level	99.5%
Dependence model	<i>t</i> -copula
<i>t</i> -copula degrees of freedom	5
Factor structure	Global + sector + region
Factor lookback period	3 years
Base currency	USD
Market data date	4 September 2026
LGD	50%

The portfolio uses USD as its reporting currency. Exposures denominated in EUR, CHF and GBP are converted into USD using the market data procedure described in Section 2.3. The final base currency exposure of the portfolio is approximately USD 899.59.

The portfolio is intentionally heterogeneous. It contains counterparties with different ratings, sectors, regions and PD estimation approaches so that migration, default, concentration and dependence effects can all be observed within a relatively small and transparent example.

5.2 Portfolio Risk Summary

The principal portfolio risk measures are shown in Table 7.

Table 7: Baseline portfolio risk results

Metric	Value (USD)
Total exposure	899.59
Default expected loss	1.17
Migration mean loss	5.14
99.5% VaR	78.31
99.5% Expected Shortfall	141.13
Unexpected loss	73.16
Maximum simulated loss	396.72
Minimum simulated loss	-18.05

The default expected loss is approximately USD 1.17, corresponding to roughly 0.13% of portfolio exposure. This measure considers the direct expected loss from counterparty default based on exposure, PD and LGD.

The mean loss generated by the full migration simulation is higher, at approximately USD 5.14. This difference is expected because the simulation includes not only default losses but

also changes in bond value caused by rating migrations and the corresponding changes in credit spreads. The simulated mean loss should therefore not be interpreted as the same quantity as the default expected loss.

At the 99.5% confidence level, portfolio VaR is approximately USD 78.31, or around 8.7% of total exposure. Expected Shortfall is materially higher at approximately USD 141.13, corresponding to roughly 15.7% of exposure. The difference between VaR and Expected Shortfall indicates that losses can increase substantially once the portfolio enters the extreme tail of the simulated distribution.

Unexpected loss, defined as VaR less the simulated mean loss, is approximately USD 73.16. This represents the portion of the VaR measure above the average simulated migration and default loss.

The maximum simulated portfolio loss is approximately USD 396.72. While this scenario is far beyond the 99.5% VaR threshold, it illustrates the severity of extreme combinations of migration and default outcomes under the dependence structure.

The minimum simulated loss is approximately USD -18.05. A negative loss represents a portfolio gain rather than a credit loss. This can occur because favorable rating migrations reduce credit spreads and increase the simulated market value of the corresponding bonds.

5.3 Exposure Concentration

Figure 3 shows the distribution of counterparty exposures after conversion into the portfolio base currency.

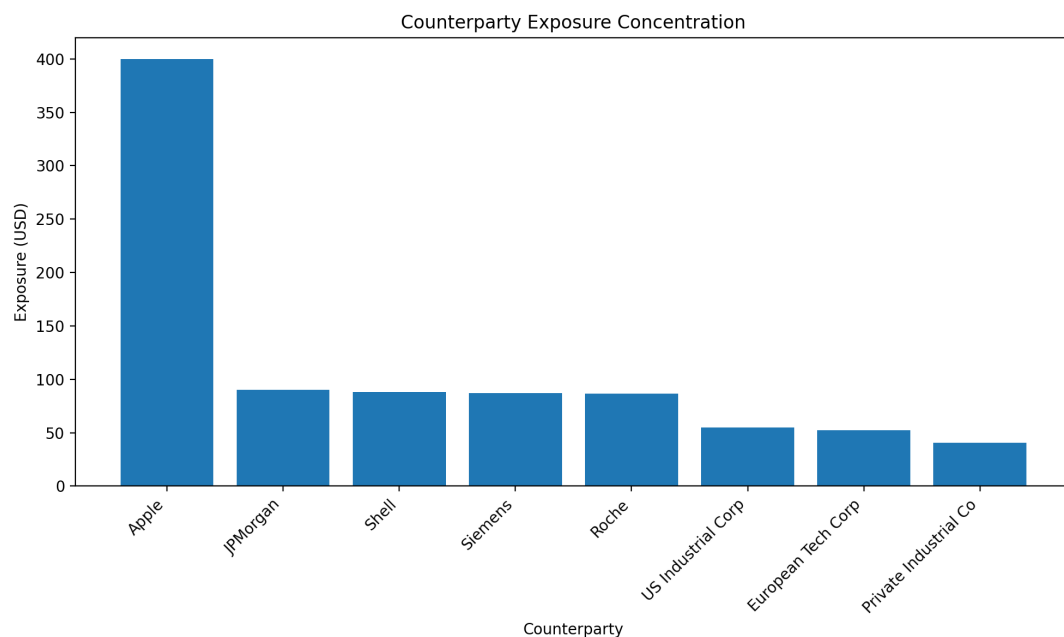


Figure 3: Counterparty exposure concentration in the baseline portfolio

The portfolio is not equally distributed across counterparties. Apple represents the largest individual position with an exposure of USD 400 and therefore accounts for approximately 44% of total portfolio exposure. The remaining positions are considerably smaller.

This concentration is relevant for the interpretation of tail risk. Even when a large counterparty has a relatively low probability of default, a sufficiently severe migration or default scenario can

produce a large absolute portfolio loss. As a result, exposure size, credit quality and dependence must be considered jointly rather than evaluated in isolation.

The importance of concentration becomes particularly visible in the Expected Shortfall measure and is investigated further in the sensitivity analysis in the following chapter.

5.4 Counterparty Credit Quality

The portfolio combines rating based and structural probabilities of default. Table 8 reports the one year PD assigned to each counterparty, together with the corresponding credit input method.

Table 8: Baseline counterparty probabilities of default

Counterparty	PD Method	Credit Input	One Year PD
Siemens	Merton	Market data calibration	4.068×10^{-13}
Apple	Rating	AA	0.0008
JPMorgan	Rating	A	0.0015
Roche	Rating	AA	0.0008
Shell	Rating	BBB	0.0040
US Industrial Corp	Rating	BBB	0.0040
European Tech Corp	Rating	BB	0.0178
Private Industrial Co	Merton	Manual calibration	0.007707

The rating based PDs follow the synthetic mapping defined in the model inputs. European Tech Corp has the highest baseline PD at 1.78%, reflecting its BB rating, while the AA counterparties Apple and Roche each have a PD of 0.08%.

The structural estimates differ more strongly. The private company example produces a materially higher PD because the manually specified equity, volatility and debt inputs describe a comparatively leveraged capital structure. By contrast, the Merton calibration for Siemens produces an extremely small model implied PD.

Figure 4 displays the same probabilities on a logarithmic scale.

The logarithmic scale is necessary because the estimated probabilities span many orders of magnitude. On a linear scale, the very small Siemens PD would be effectively invisible relative to the higher rating based and private company probabilities.

The extremely low Siemens PD illustrates an important limitation of the pure Merton framework. When estimated firm asset value is high relative to the debt default point and asset volatility is moderate, the implied distance to default can become very large, leading to a probability of default close to zero. Such an estimate should therefore be interpreted as a structural model output rather than as a complete assessment of real world credit risk.

Importantly, a very low structural default PD does not imply that the counterparty cannot contribute to portfolio risk. The migration framework also captures non-default rating transitions and associated spread revaluation. Consequently, Siemens can generate nonzero VaR and Expected Shortfall contributions even though its structural default probability is extremely small.

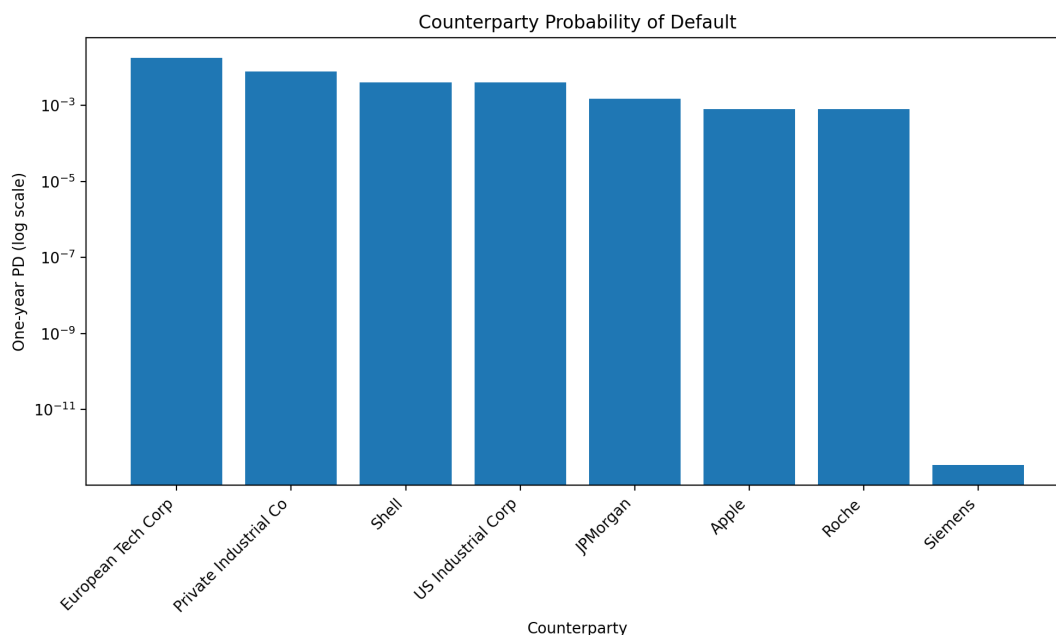


Figure 4: Counterparty probabilities of default in the baseline portfolio

5.5 Simulated Loss Distribution

Figure 5 shows the complete simulated portfolio loss distribution.

The distribution is strongly asymmetric. A large proportion of scenarios generate relatively small losses or moderate changes in portfolio value, while a much smaller number of scenarios produce substantially larger losses.

This asymmetry is a natural consequence of credit portfolio risk. Rating upgrades and moderate downgrades generally generate limited valuation changes, whereas severe downgrades and defaults can generate losses that are large relative to the initial expected loss.

The distribution also contains negative observations because favorable migration scenarios can increase bond values. The positive tail, however, extends considerably further than the negative side of the distribution, producing the characteristic right-skewed loss profile.

The difference between the simulated mean loss, VaR and Expected Shortfall is clearly visible. VaR identifies the loss threshold corresponding to the selected confidence level, whereas Expected Shortfall reflects the average severity of the scenarios beyond that threshold.

5.6 Tail Losses

The extreme portion of the simulated loss distribution is shown separately in Figure 6.

The tail view highlights the difference between the 99.5% VaR of approximately USD 78.31 and the corresponding Expected Shortfall of approximately USD 141.13. Once losses exceed the VaR threshold, their severity increases considerably.

Large counterparty exposures create discrete loss concentrations in the tail, particularly when severe migration states or defaults occur. Dependence between counterparties further increases the possibility that unfavorable credit outcomes occur simultaneously.

This behavior explains why Expected Shortfall is especially informative for the portfolio. VaR

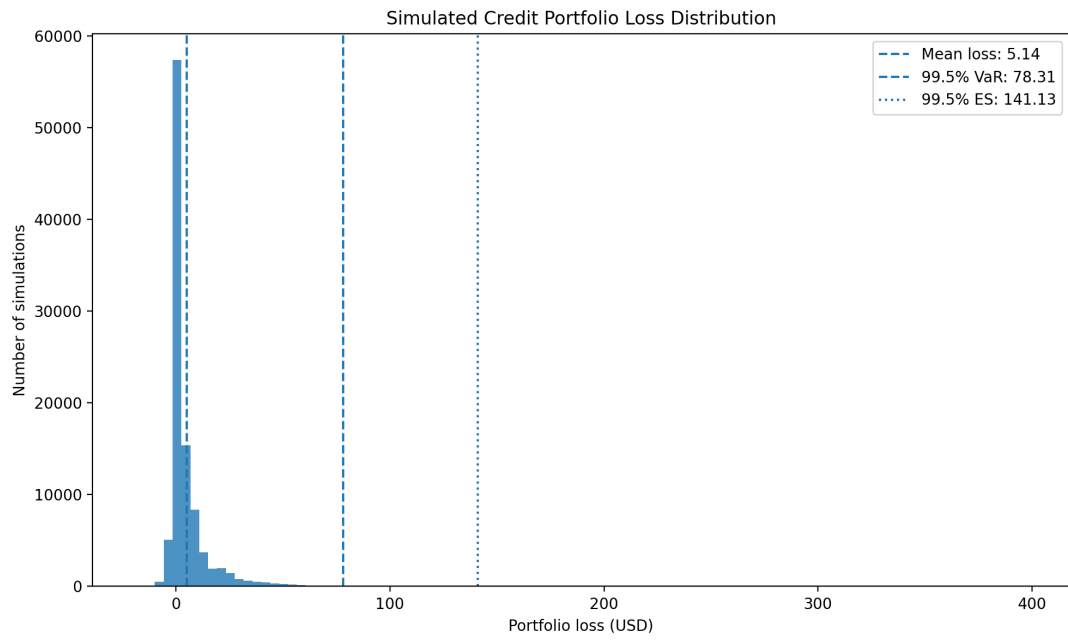


Figure 5: Simulated baseline credit portfolio loss distribution

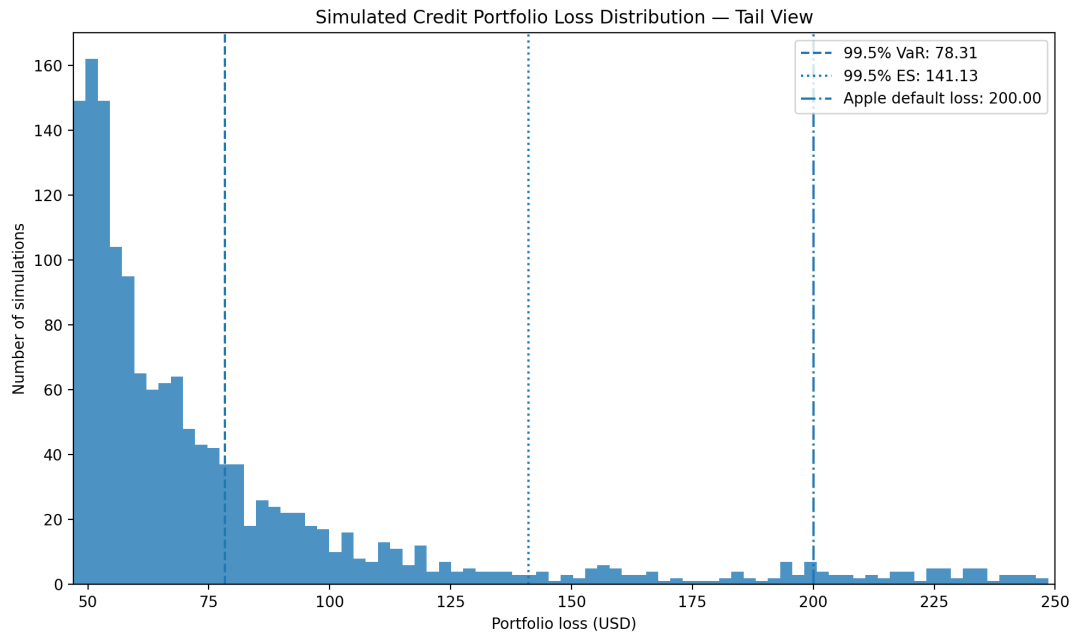


Figure 6: Tail of the simulated baseline credit portfolio loss distribution

identifies the location of a high loss quantile, but it does not describe the magnitude of losses beyond that point. Expected Shortfall captures this additional tail severity.

5.7 Counterparty Risk Contributions

Figure 7 compares the incremental VaR and incremental Expected Shortfall associated with the individual counterparties.

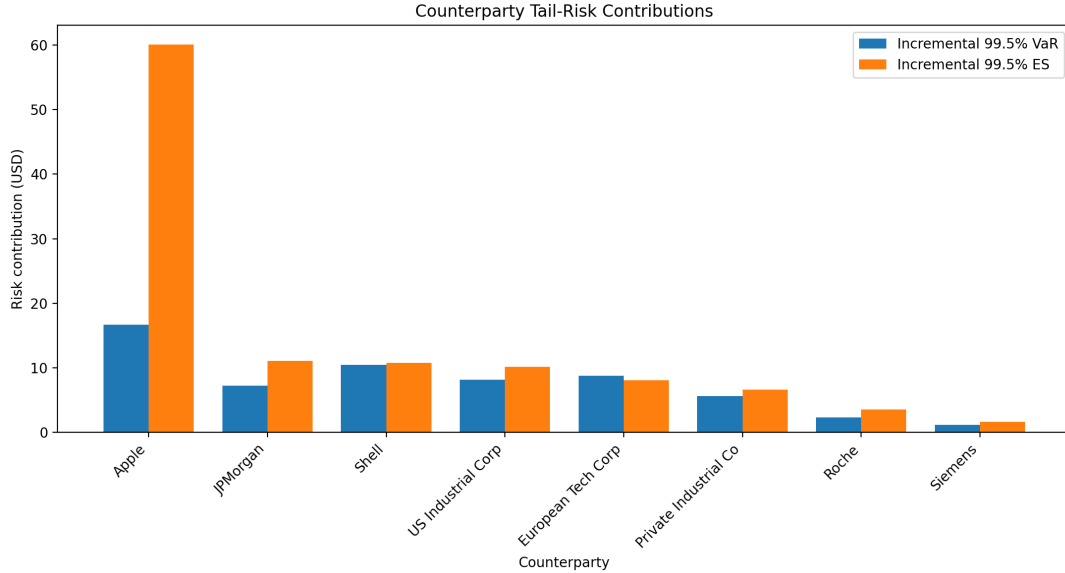


Figure 7: Counterparty incremental VaR and Expected Shortfall

A counterparty's contribution to portfolio risk depends on several characteristics simultaneously: exposure size, probability of migration or default, LGD, sensitivity to changes in credit spreads and dependence with the remainder of the portfolio. Consequently, the largest exposure does not necessarily dominate every risk measure, and counterparties with relatively low default probabilities can still contribute materially to portfolio tail risk.

The distinction between default risk and migration risk is particularly important for the structural Merton counterparties. Siemens, for example, has an extremely small Merton implied probability of default in the baseline calibration. Nevertheless, it still exhibits a nonzero contribution to portfolio VaR and Expected Shortfall.

This result is not primarily driven by the structural default probability. For the migration simulation, the structural PD is translated into an equivalent rating category and the counterparty is subsequently exposed to the same migration mechanism as the rating based counterparties. A downgrade is associated with a higher credit spread and therefore a higher discount rate applied to the remaining bond cash flows. This reduces the bond's present value and generates a mark-to-market loss. Conversely, an upgrade reduces the credit spread, lowers the required discount rate and can increase the bond value, resulting in a negative simulated loss.

The Siemens result therefore illustrates an important feature of the framework: a very small default PD does not imply negligible total credit risk. Structural default risk, rating migration risk and spread revaluation represent related but distinct channels through which a counterparty may affect the portfolio loss distribution. Siemens can therefore generate nonzero portfolio losses even though its structural probability of default is extremely small.

Systematic dependence can further strengthen these effects. Under the baseline t -copula speci-

cation, adverse migration outcomes can occur jointly across several counterparties when common global, regional or sector factors deteriorate. The t -copula therefore amplifies the severity of some joint tail scenarios. However, for counterparties such as Siemens, the existence of a nonzero risk contribution is primarily explained by the migration and spread revaluation mechanism rather than by its extremely small Merton default probability.

The difference between incremental VaR and incremental Expected Shortfall provides additional information about the shape of a counterparty's contribution to the portfolio tail. A counterparty may have a moderate effect around the VaR quantile but a considerably larger impact in the most severe scenarios captured by Expected Shortfall.

The incremental measures shown here are calculated using leave one out portfolio differences. They should therefore not be interpreted as an additive Euler allocation of total VaR or Expected Shortfall. The individual incremental contributions do not generally sum to the corresponding total portfolio risk measure.

5.8 Baseline Interpretation

The baseline results illustrate several important features of the framework and show why portfolio credit risk cannot be understood from probabilities of default alone.

First, default expected loss represents only a relatively small part of the total simulated credit risk. The default expected loss of approximately USD 1.17 reflects only exposure, PD and LGD. Once rating migration and the associated spread revaluation are included, the mean simulated portfolio loss rises to approximately USD 5.14. The difference demonstrates that credit deterioration short of default can be an important source of economic loss.

Second, the large difference between VaR and Expected Shortfall highlights the asymmetric nature of the portfolio loss distribution. The 99.5% VaR is approximately USD 78.31, while the corresponding Expected Shortfall is approximately USD 141.13. Losses beyond the VaR threshold are therefore substantially more severe than the quantile itself suggests. This illustrates why Expected Shortfall provides important additional information for portfolios exposed to rare but large credit events.

Third, exposure concentration remains relevant even when the largest counterparty has high credit quality. A low PD reduces the frequency of severe credit outcomes but does not eliminate the loss severity associated with a large exposure. This becomes particularly important in the extreme tail, where a rare deterioration or default of a large counterparty may have a disproportionate effect on portfolio losses.

Fourth, the Siemens example demonstrates the difference between default probability and total portfolio credit risk. Its structural Merton PD is extremely small, yet the counterparty still contributes to VaR and Expected Shortfall because the portfolio model also captures rating migration and spread revaluation. The equivalent rating used for the migration simulation creates a pathway through which non-default credit deterioration affects portfolio value even when structural default risk is negligible.

This distinction is important when interpreting structural credit models. A very low Merton PD should not be interpreted as evidence that a counterparty is economically risk free. It reflects the probability of crossing the structural default boundary under the assumptions of the Merton model. Other sources of credit risk, including rating deterioration, changes in credit spreads, concentration and systematic dependence, remain relevant.

Finally, the baseline results demonstrate the interaction between counterparty specific credit quality and systematic dependence. Individual PDs and migration probabilities determine the

unconditional likelihood of adverse credit states, while common global, regional and sector factors influence the extent to which such outcomes occur jointly. The t -copula adds additional tail dependence, increasing the probability of simultaneously adverse latent credit conditions relative to an independent framework.

The baseline portfolio should therefore be interpreted as the combined result of several interacting mechanisms: counterparty exposure, PD, LGD, rating migration, spread revaluation, concentration and systematic dependence. No single input explains the portfolio tail in isolation.

The baseline case provides the reference configuration for the following validation analysis, where the main assumptions are challenged individually through alternative dependence structures, t -copula degrees of freedom, LGD assumptions, concentration scenarios, factor structures, confidence levels and Monte Carlo stability tests.

6 Model Validation and Sensitivity Analysis

The baseline results presented in Section 5 depend on a number of modeling assumptions and numerical choices. A dedicated validation framework is therefore used to challenge the principal drivers of portfolio risk and to evaluate the numerical stability of the Monte Carlo implementation.

The validation analysis covers the dependence model, the degrees of freedom of the Student t -copula, LGD, portfolio concentration, systematic factor structure, Monte Carlo sample size and random seed, and the confidence level used for tail risk measurement.

Unless explicitly varied in a particular experiment, the validation runs use the baseline configuration of 100,000 simulations, random seed 0, a 99.5% confidence level, a Student t -copula with five degrees of freedom, the global, sector and region factor structure, and an LGD of 50%. The configured market data date for FX rates and risk free rates is fixed at 4 September 2026.

The experiments are constructed so that the parameter under investigation is changed while the remaining relevant assumptions are kept fixed. Automated integrity checks verify the stored model settings and simulation outputs before the validation statistics are calculated. The final validation run passed all experiment integrity checks.

6.1 Dependence Model Comparison

The first experiment examines the effect of alternative assumptions regarding dependence between counterparties. Three specifications are compared: independent credit outcomes, a Gaussian copula and the baseline Student t -copula with five degrees of freedom.

Table 9: Dependence model comparison

Dependence model	Mean loss	99.5% VaR	99.5% ES
Independent	5.08	49.92	100.27
Gaussian copula	5.11	56.28	108.15
Student t -copula, $df = 5$	5.14	78.31	141.13

The mean simulated loss changes only modestly across the three dependence specifications because the marginal credit risk of the individual counterparties is unchanged. Tail risk, however, changes substantially.

Relative to the independent specification, the baseline Student t-copula increases the 99.5% VaR from approximately USD 49.92 to USD 78.31 and Expected Shortfall from USD 100.27 to USD 141.13. The Gaussian copula produces an intermediate result.

This illustrates the distinction between marginal credit risk and portfolio dependence. Dependence has a relatively limited effect on the center of the loss distribution but a much stronger effect on the probability and severity of scenarios in which several counterparties experience adverse credit outcomes simultaneously.

The Student t-copula produces the highest tail risk because it allows stronger dependence in extreme latent factor realizations than the Gaussian specification. This is consistent with the tail dependence mechanism described in the methodology section.

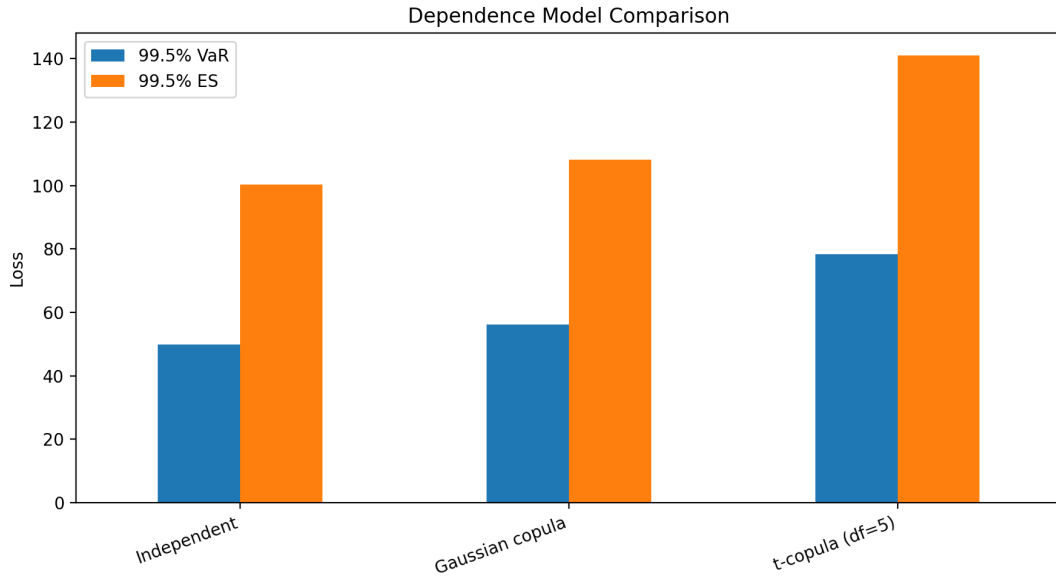


Figure 8: Sensitivity of portfolio tail risk to the dependence model

6.2 Student t-Copula Degrees of Freedom

The second experiment investigates the sensitivity of the Student t-copula to its degrees of freedom parameter. Values of 3, 5 and 10 are compared.

Table 10: Sensitivity to Student t-copula degrees of freedom

Degrees of freedom	Mean loss	99.5% VaR	99.5% ES
3	5.10	87.12	156.65
5	5.14	78.31	141.13
10	5.16	70.10	130.67

The results show a clear relationship between the degrees of freedom and portfolio tail risk. Lower degrees of freedom imply heavier tails and stronger tail dependence. Consequently, the most severe specification, with three degrees of freedom, produces a 99.5% VaR of approximately USD 87.12 and Expected Shortfall of USD 156.65.

When the degrees of freedom increase to 10, VaR falls to approximately USD 70.10 and Expected Shortfall to USD 130.67. The change is economically meaningful even though the mean simulated

loss remains close to USD 5 across all three cases.

This experiment demonstrates that the degrees of freedom parameter is an important model assumption. It has limited influence on the center of the loss distribution but a substantial influence on the extreme tail. The baseline choice of five degrees of freedom therefore represents a meaningful modeling decision rather than a purely technical parameter.

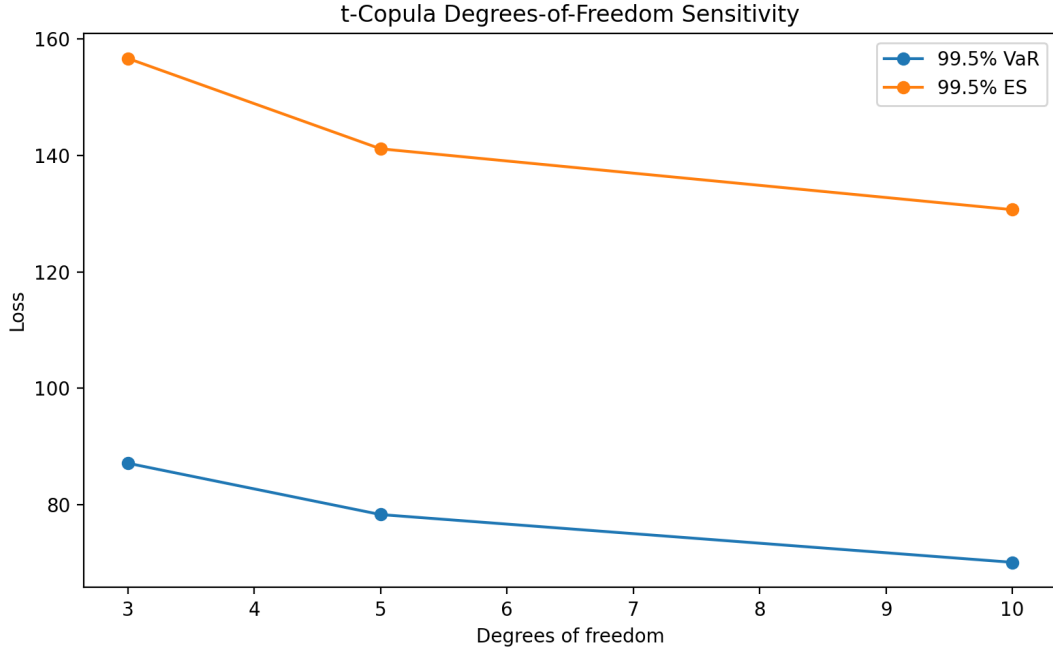


Figure 9: Sensitivity of portfolio tail risk to Student t-copula degrees of freedom

6.3 LGD Sensitivity

The LGD sensitivity analysis applies deterministic LGDs of 40%, 50% and 60% to all counterparties while keeping the remaining baseline assumptions unchanged.

Table 11: LGD sensitivity

LGD	Default EL	Mean loss	99.5% VaR	99.5% ES
40%	0.94	4.90	70.06	126.02
50%	1.17	5.14	78.31	141.13
60%	1.40	5.39	86.52	156.94

Default expected loss increases directly with LGD because default loss is proportional to exposure, PD and LGD.

The effect on the complete simulated distribution is also monotonic. Increasing LGD from 40% to 60% raises VaR from approximately USD 70.06 to USD 86.52 and Expected Shortfall from USD 126.02 to USD 156.94.

The relationship is not limited to expected loss because LGD also determines the severity of default outcomes within the Monte Carlo loss distribution. The impact becomes particularly important in tail scenarios involving one or several large counterparty defaults.

The experiment confirms that the deterministic LGD assumption is an important source of model sensitivity. In a more advanced framework, recovery uncertainty or stochastic LGD could introduce an additional source of tail risk beyond the deterministic specification used here.

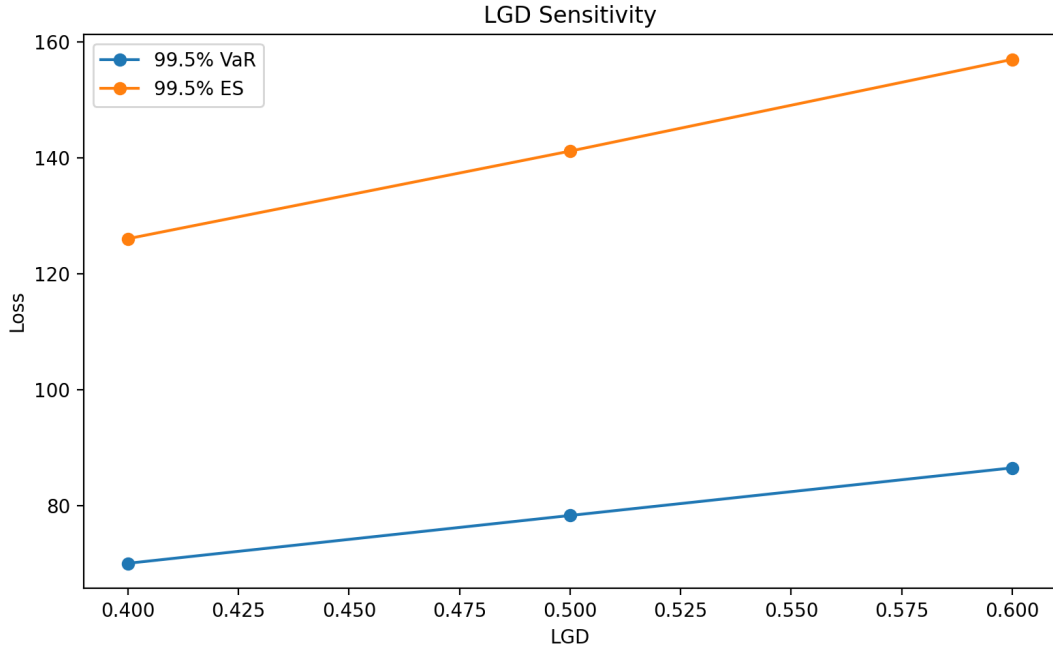


Figure 10: Sensitivity of portfolio risk to LGD

6.4 Portfolio Concentration Sensitivity

The concentration experiment changes the exposure to Apple while proportionally rescaling the remaining counterparty exposures so that total portfolio exposure in the base currency remains approximately constant.

The experiment considers Apple exposures of USD 100, USD 400 and USD 800. The corresponding portfolio shares and Herfindahl–Hirschman Index (HHI) values are shown together with the resulting risk measures in Table 12.

Table 12: Portfolio concentration sensitivity

Apple	Share	HHI	Default EL	99.5% VaR	99.5% ES
100	11.12%	0.134	1.66	103.03	138.01
400	44.46%	0.245	1.17	78.31	141.13
800	88.93%	0.793	0.52	49.03	182.26

The results require careful interpretation. Increasing the Apple exposure does not produce a monotonic increase in every risk measure. Default expected loss and VaR decline as the portfolio becomes increasingly concentrated in Apple.

This occurs because the experiment changes not only concentration but also the credit quality composition of the portfolio. Apple has a relatively low PD in the demonstration portfolio. Real-locating exposure from higher PD counterparties toward Apple therefore lowers default expected loss and reduces the frequency of sufficiently large losses around the 99.5% VaR quantile.

Expected Shortfall shows a different effect. At the most concentrated configuration, where Apple represents approximately 89% of total portfolio exposure, Expected Shortfall rises to approximately USD 182.26 despite the substantially lower VaR of USD 49.03.

A rare severe migration or default of the dominant counterparty produces a very large portfolio loss. Such outcomes can remain sufficiently infrequent that the VaR quantile falls, while still contributing strongly to the average loss among the most severe tail scenarios. Expected Shortfall therefore captures a concentration effect that is not visible from VaR alone.

The experiment illustrates both single counterparty concentration risk and the different information contained in VaR and Expected Shortfall. It should not, however, be interpreted as a pure concentration experiment with constant credit quality. Changing Apple's weight simultaneously changes the credit quality composition of the portfolio.

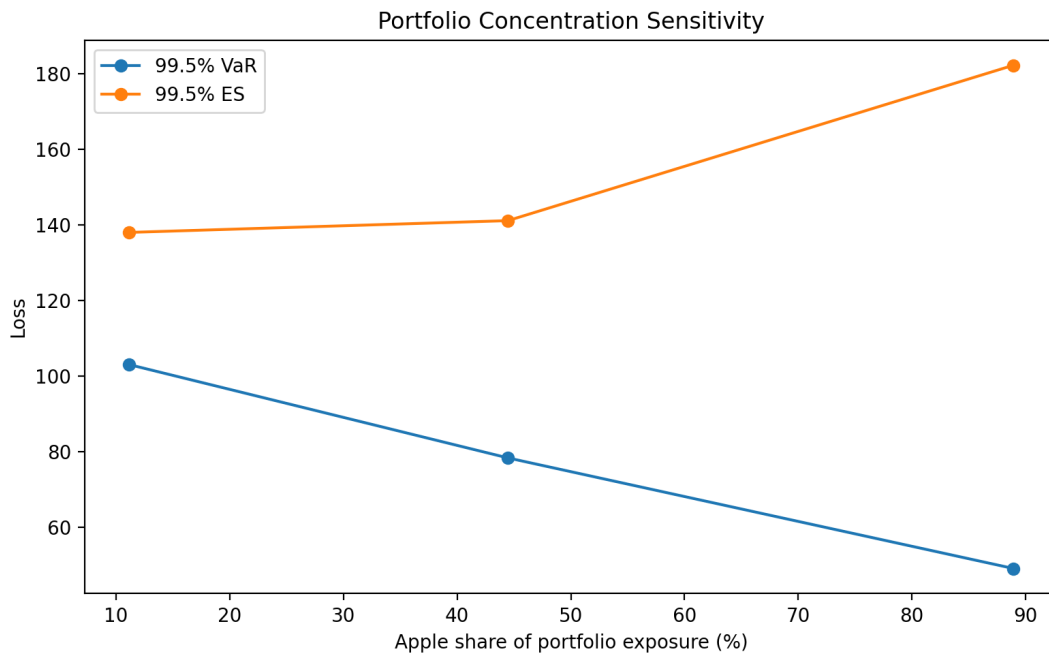


Figure 11: Sensitivity of portfolio risk to counterparty concentration

6.5 Factor Structure Sensitivity

The factor structure sensitivity compares four specifications: a single global factor, global and sector factors, global and region factors, and the complete global, sector and region structure used in the baseline model.

Table 13: Systematic factor structure sensitivity

Factor structure	Mean loss	99.5% VaR	99.5% ES
Single factor	5.07	76.29	133.75
Global + sector	5.03	73.91	130.24
Global + region	5.05	74.16	139.13
Global + sector + region	5.14	78.31	141.13

The factor structure affects portfolio risk because it changes the channels through which systematic movements are transmitted across counterparties.

The differences are smaller than those generated by changing the copula specification or its degrees of freedom, but they remain economically relevant. In this demonstration portfolio, the complete global, sector and region specification produces the highest VaR and Expected Shortfall.

This result should not be interpreted as evidence that a more detailed factor structure must always produce higher risk. The outcome depends on the estimated counterparty loadings, the selected factor proxies and the composition of the portfolio. The comparison instead demonstrates that systematic factor specification is a meaningful source of model risk and should be challenged rather than treated as mechanically fixed.

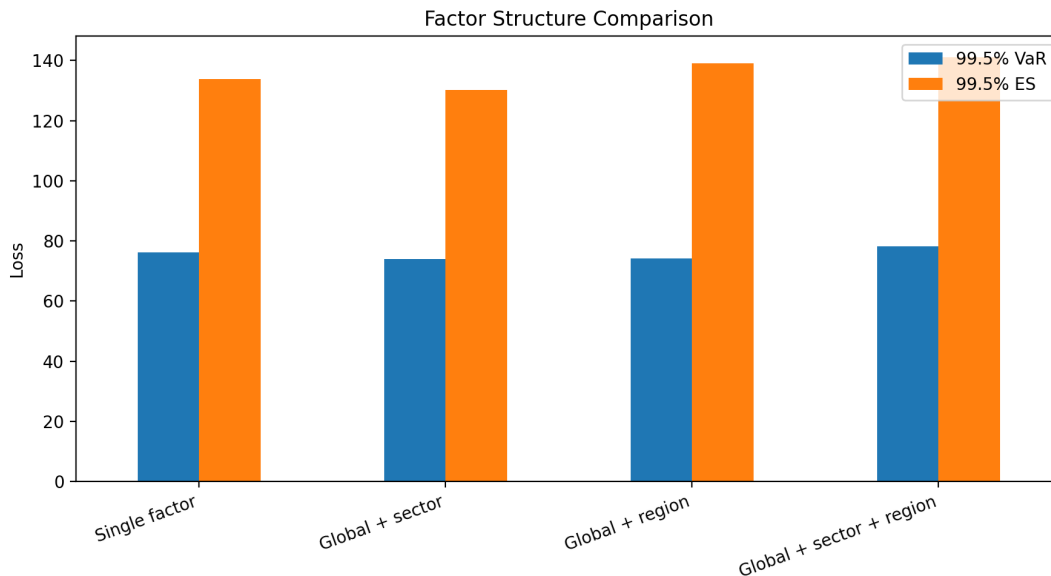


Figure 12: Sensitivity of portfolio risk to the systematic factor structure

6.6 Monte Carlo Stability

Monte Carlo risk measures contain simulation error because they are estimated from a finite number of scenarios. This is particularly relevant for high confidence VaR and Expected Shortfall because only a small fraction of the simulated observations determines the extreme tail.

To assess numerical stability, the model is run using 10,000, 50,000, 100,000 and 250,000 scenarios. For each simulation size, three random seeds are used. All portfolio and model assumptions are otherwise held fixed.

Table 14 summarizes the average VaR and Expected Shortfall across the three seeds together with their relative standard deviations.

Table 14: Monte Carlo stability across random seeds

Simulations	Mean VaR	VaR rel. std.	Mean ES	ES rel. std.
10,000	75.23	8.39%	131.61	8.09%
50,000	76.22	3.21%	133.18	0.27%
100,000	76.47	2.08%	137.71	2.60%
250,000	75.79	0.33%	135.49	1.00%

The 10,000 scenario configuration exhibits substantial variation across seeds, with relative standard deviations of approximately 8.4% for VaR and 8.1% for Expected Shortfall. This indicates that the smallest simulation sample produces relatively unstable tail estimates for this analysis.

The dispersion is substantially lower at larger simulation sizes. At 250,000 simulations, the relative standard deviation is approximately 0.33% for VaR and 1.00% for Expected Shortfall, indicating considerably greater numerical stability.

The reduction is not perfectly monotonic for every statistic. In particular, the three 50,000 scenario runs happen to produce very similar Expected Shortfall estimates, while the 100,000 scenario runs exhibit greater dispersion. This should not be interpreted as evidence that 50,000 simulations are intrinsically more accurate. Only three random seeds are used at each simulation size, meaning that the estimated dispersion across seeds is itself subject to sampling variation.

The experiment nevertheless shows that very small simulation samples can produce materially unstable tail estimates and that larger simulation samples improve overall numerical robustness. The baseline choice of 100,000 simulations represents a practical compromise between computational effort and stability for the demonstration framework, while 250,000 simulations provide the strongest stability among the tested configurations.

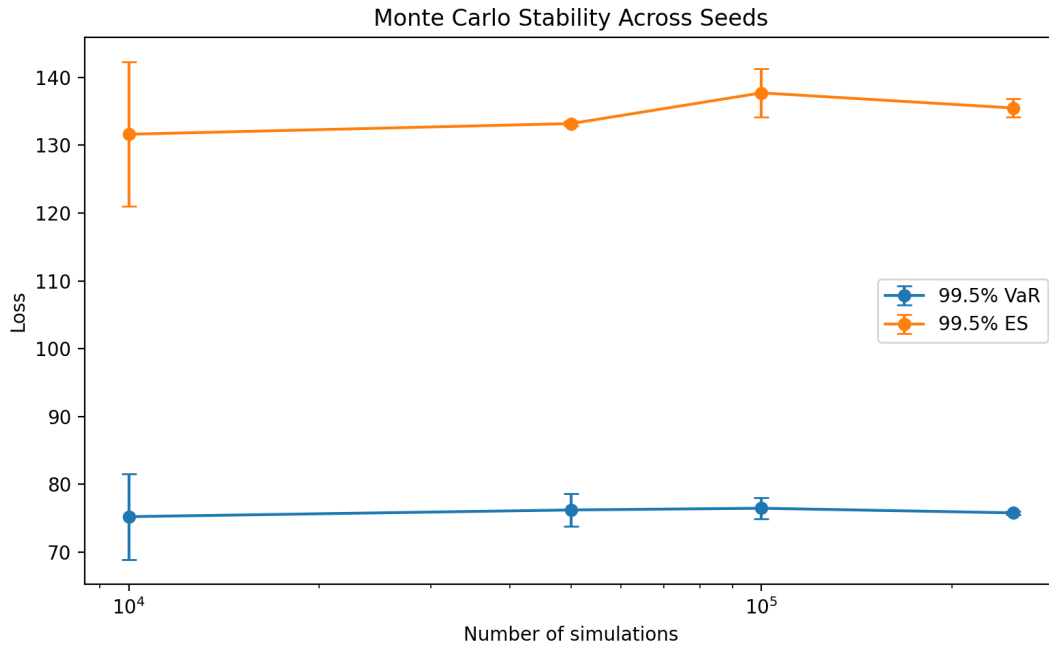


Figure 13: Monte Carlo stability across simulation sizes and random seeds

6.7 Confidence Level Sensitivity

The confidence level sensitivity is calculated from the same 100,000 baseline loss scenarios. This isolates the effect of changing the point in the loss distribution at which VaR and Expected Shortfall are measured.

Table 15: Confidence level sensitivity

Confidence	VaR	Expected Shortfall	Tail scenarios
95.0%	25.28	49.55	5,000
99.0%	56.93	103.59	1,000
99.5%	78.31	141.13	500
99.9%	202.39	256.57	100

Both risk measures increase strongly as the confidence level moves further into the tail. VaR increases from approximately USD 25.28 at 95% to USD 202.39 at 99.9%, while Expected Shortfall rises from approximately USD 49.55 to USD 256.57.

The number of simulated observations used to estimate the tail decreases rapidly as the confidence level increases. At the baseline confidence level of 99.5%, the Expected Shortfall calculation uses 500 tail observations. At 99.9%, only 100 observations remain.

This highlights an important interaction between confidence level and simulation size. Higher confidence levels provide information about increasingly extreme events but also increase sampling uncertainty because fewer simulated observations are available to estimate the relevant part of the loss distribution.

As an additional implementation check, VaR and Expected Shortfall were recalculated independently from the stored baseline scenario losses at the 99.5% confidence level. Both values reproduced the corresponding core model results exactly, with zero difference in the validation calculation.

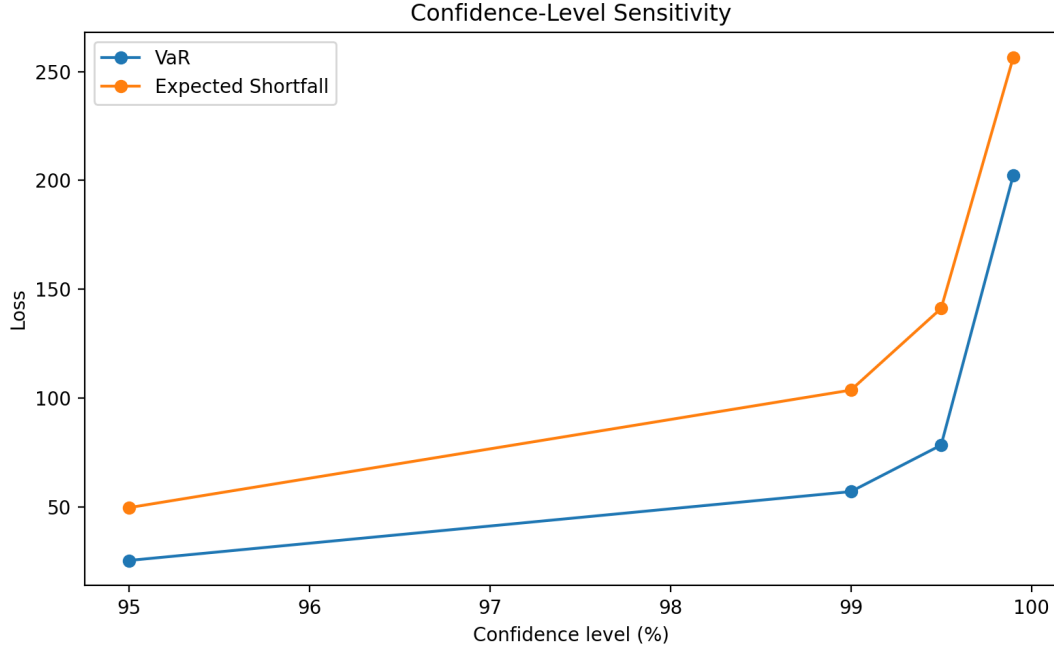


Figure 14: Sensitivity of VaR and Expected Shortfall to the confidence level

6.8 Overall Validation Assessment

The validation results show that the model responds consistently to changes in its principal assumptions and numerical settings.

Dependence assumptions are among the strongest drivers of portfolio tail risk. Moving from independent outcomes to Gaussian dependence and subsequently to the Student t-copula produces substantially larger VaR and Expected Shortfall while leaving the mean portfolio loss comparatively stable. The degrees of freedom sensitivity further shows that stronger tail dependence materially increases extreme portfolio losses.

LGD affects both expected and tail losses in the expected direction, while the factor structure has a smaller but still meaningful influence on portfolio risk. The concentration experiment demonstrates that portfolio risk cannot be assessed from exposure concentration alone: credit quality composition influences the frequency of losses, while severe single counterparty outcomes can dominate Expected Shortfall.

The Monte Carlo analysis confirms that tail estimates are sensitive to simulation size and random seed when the number of scenarios is small. Larger simulation samples reduce overall sampling variability, although the limited number of seeds means that the observed reduction is not perfectly monotonic. The confidence level analysis further demonstrates that increasingly extreme risk measures are estimated from progressively fewer tail observations.

No individual validation experiment establishes that the model is correct in an absolute sense. Instead, the combined evidence shows that the implementation responds coherently to changes in its principal assumptions, that the main sources of model sensitivity are identifiable, and that the baseline results can be reproduced and challenged systematically.

The validation therefore identifies several assumptions that require particular attention when interpreting the model results: dependence structure, Student t-copula degrees of freedom, LGD, portfolio concentration, systematic factor specification, and the numerical precision of extreme tail estimates. These issues, together with further structural simplifications of the framework, are discussed in the following model limitations section.

7 Model Limitations

The framework combines several credit risk components within a transparent and testable implementation, but it remains a demonstration model rather than a production credit risk system. The results should therefore be interpreted in the context of the assumptions made at each stage of the modeling process.

The main limitations arise from the demonstration data, structural probability of default estimation, rating migration assumptions, spread based revaluation, factor and copula dependence structure, deterministic recovery assumptions, market data treatment, Monte Carlo estimation, and the interpretation of counterparty risk contributions.

The sensitivity analysis in Section 6 provides evidence on several of these assumptions, but sensitivity analysis does not eliminate model risk. It shows how the implementation responds when assumptions are changed rather than establishing that the assumptions themselves are empirically correct.

7.1 Demonstration Portfolio and Scope

The portfolio is intentionally small and contains only eight counterparties. Exposures, ratings, LGDs, bond characteristics and several other inputs are synthetic demonstration values. The portfolio is designed to illustrate model mechanics rather than reproduce the composition of an actual institutional credit portfolio.

Consequently, the resulting risk measures should not be interpreted as estimates of the risk of the named companies or as a realistic calibration of a particular investment portfolio. The presence of real listed counterparties is used primarily to demonstrate market data retrieval, structural calibration and factor estimation.

The framework also assumes that counterparty exposures are fixed over the one year simulation horizon. It does not simulate changes in exposure at default, loan drawdowns, collateral, netting, guarantees, contractual optionality or future exposure from derivatives. It should therefore not be interpreted as a complete counterparty credit exposure or CVA framework.

Similarly, the framework does not explicitly distinguish between different levels of debt seniority, security, collateralization or legal recovery structure. These characteristics can materially affect actual loss severity and recovery behavior.

7.2 Structural Merton Model

The Merton model provides an economically intuitive connection between a firm's capital structure, equity value, market volatility and default risk [9]. However, this transparency is obtained through strong structural assumptions.

The implementation follows the standard Merton representation in which firm asset value is modeled using a continuous diffusion process and equity is interpreted as a call option on firm assets. Asset volatility and the risk free rate are treated as constant over the selected horizon, while default is determined by comparing firm asset value with a single debt barrier at the end of the horizon.

Real corporate asset values can exhibit jumps, changing volatility, nonlinear capital structures and sudden deterioration. Default can also occur before the selected horizon. These effects are not captured by the standard terminal default structure.

The debt side of the calibration is necessarily simplified. A firm's actual liability structure may contain short and long term debt, multiple seniorities, revolving facilities, leases, contingent liabilities and different maturity profiles. Reducing these obligations to a single default barrier introduces model approximation.

Firm asset value and asset volatility are themselves latent quantities and must be inferred numerically from equity information. Errors or timing differences in equity value, equity volatility, debt information or the selected risk free rate therefore propagate into the calibrated asset parameters and ultimately into the structural PD.

A particularly visible limitation is the possibility of extremely small model implied PDs. A company with a large market capitalization relative to its estimated debt barrier and moderate equity volatility can obtain a very large distance to default. The resulting PD can then approach zero even though the company is not economically risk free.

This effect appears directly in the demonstration portfolio for Siemens. The very small structural PD should therefore be interpreted as an output of the implemented Merton assumptions rather than as a complete assessment of real world creditworthiness.

The implemented PD also uses the risk free rate within the standard Merton equations. It is therefore best interpreted as a structural model implied quantity. It has not been statistically calibrated to reproduce observed long run default frequencies and should not automatically be interpreted as an empirical physical probability of default.

For market based Merton calibration, equity information and balance sheet information may also refer to different observation dates. Equity markets react continuously, whereas accounting

debt information is updated much less frequently. This creates an additional timing mismatch between the inputs entering the structural calibration.

7.3 Rating Mapping and Credit Migration

The rating based PD mapping and the one year rating migration matrix are synthetic model inputs. They do not reproduce the historical transition experience or PD methodology of a particular rating agency.

The same transition structure is applied without conditioning on the business cycle, industry, geography, issuer characteristics or previous rating history. In practice, migration probabilities can vary materially between normal and stressed periods and may exhibit rating momentum, sector effects and regime dependence.

The transition matrix is also static over the one year simulation horizon. The framework does not model a sequence of migrations through time. For example, a counterparty cannot experience an intermediate downgrade followed by another deterioration before the final horizon. The model instead produces one end of horizon rating state.

For counterparties using the Merton approach, the continuous structural PD must be translated into an equivalent discrete rating category before the migration simulation can be applied. This creates a common migration framework across rating and Merton counterparties, but it also introduces an additional approximation.

A continuous structural PD contains more granular information than a discrete rating. After conversion into an equivalent rating, the migration probabilities are determined by the corresponding transition matrix row. The exact Merton PD is therefore not preserved continuously inside the migration mechanism. Small changes in structural PD can also lead to a change in equivalent rating when a rating boundary is crossed.

This approximation is particularly important when interpreting counterparties such as Siemens. Its extremely small structural PD does not directly imply an equally small migration loss contribution because the equivalent rating subsequently exposes the counterparty to non-default migration states and the associated spread revaluation.

7.4 Credit Spread and Bond Revaluation

Non-default migration losses are represented through rating dependent credit spreads. A downgrade increases the applicable spread and therefore the discount rate used for the remaining bond cash flows, reducing the present value of the position. An upgrade has the opposite effect and can generate a mark-to-market gain.

This mechanism captures the basic economic relationship between credit quality and bond valuation, but the spread representation is simplified. The model assigns synthetic spreads by rating category rather than using counterparty specific market credit spreads.

Actual credit spreads depend not only on rating but also on maturity, seniority, liquidity, sector, currency, market conditions, issuer specific information and the broader credit cycle. Two counterparties with the same rating can therefore trade at substantially different spreads.

The framework does not model a complete credit spread term structure. Similarly, the risk free input is not a fully calibrated term structure for every bond maturity. Consequently, the revaluation should be interpreted as a simplified migration mark-to-market approximation rather than a full market consistent bond valuation engine.

Spread changes within a rating category are also absent. A counterparty can experience a substantial market spread widening without an immediate formal rating downgrade, but such changes are not represented unless the simulated rating state changes.

The migration loss is a mark-to-market effect. It should therefore be distinguished from a realized credit loss under a hold-to-maturity strategy. Conversely, default is treated separately through exposure multiplied by LGD rather than by explicitly modeling recovery cash flows and their timing.

7.5 LGD and Exposure at Default

LGD is deterministic in the baseline portfolio and is set to 50% for every counterparty. The sensitivity analysis demonstrates that changing this assumption has a material effect on VaR and Expected Shortfall.

In practice, recovery rates differ by issuer, seniority, collateral, industry, jurisdiction and macroeconomic conditions. Recovery can also be systematically lower during periods in which default rates are high.

The current framework does not model such dependence between probability of default, systematic stress and LGD. In particular, severe common credit conditions can increase the number of defaults in the simulation without simultaneously reducing recovery rates.

This may underrepresent an important source of tail risk because default frequency and loss severity can deteriorate together during stressed periods.

Exposure is also deterministic over the simulation horizon. The model therefore does not capture adverse dependence between exposure at default and counterparty credit quality, commonly referred to as wrong way risk.

7.6 Systematic Factor Model

Systematic dependence is represented using global, regional and sector factors together with an idiosyncratic component. Factor loadings are estimated from historical equity returns.

Equity returns provide a convenient observable source for estimating systematic sensitivities, but equity market dependence is not identical to dependence in credit migration or default. The historical relationship between an equity return and a market or sector index therefore provides only a proxy for the corresponding latent credit dependence.

The estimated factor loadings are assumed to remain stable over the simulation horizon. In reality, systematic sensitivities can vary over time, particularly during periods of market stress, corporate restructuring or changes in business composition.

The selected global, regional and sector proxies are also modeling choices. Different indices, lookback periods or data frequencies can produce different estimated loadings. Some market indices overlap economically and statistically, which means that the decomposition into global, regional and sector components should not be interpreted as a unique structural representation of the underlying economic risks.

For listed counterparties, the model uses the company's own equity ticker where possible. For unlisted or synthetic counterparties, a listed proxy ticker is used. This introduces proxy risk because the selected company may differ from the modeled counterparty in leverage, business model, geography, size or sensitivity to systematic risk.

The factor source is recorded in the model output, which improves transparency, but it does not remove the economic uncertainty associated with proxy selection.

7.7 Copula and Dependence Assumptions

The validation results demonstrate that the dependence specification is one of the most important drivers of portfolio tail risk. This sensitivity is itself a major source of model risk.

The Gaussian copula does not generate the same degree of extreme joint dependence as the Student t-copula. The Student t-copula addresses this limitation by introducing heavier joint tails, but the degrees of freedom parameter becomes an important model assumption.

The baseline value of five degrees of freedom is a configurable modeling choice and is challenged through sensitivity analysis. It has not been estimated from a sufficiently large portfolio of observed migration and default data within this project. The validation results show that alternative values can materially change VaR and Expected Shortfall.

The Student t-copula also imposes a particular parametric form of dependence. In particular, its tail dependence is symmetric in the underlying latent distribution, whereas actual credit dependence may behave differently during favorable and adverse economic conditions [12].

More generally, the dependence model is parsimonious. It captures shared global, regional and sector effects but does not attempt to reproduce every pairwise counterparty relationship, nonlinear interaction, contagion mechanism or direct default linkage.

The factor structure and the copula should therefore be interpreted jointly as a stylized dependence representation rather than as a uniquely identified description of the true joint credit distribution.

7.8 Market Data and Currency Treatment

The framework automatically retrieves several market inputs and retains source indicators for FX rates and risk free rates. Nevertheless, external market data introduce availability, timing and data quality risk.

The configured `market_data_date` applies specifically to FX rates and risk free rates. It should not be interpreted as a complete historical valuation date for all model inputs.

Equity information used for market based Merton calibration and historical equity data used for factor estimation retain their own retrieval procedures and rolling historical windows. As a result, setting an historical `market_data_date` does not create a fully point in time reconstruction of the complete portfolio model.

The source used for FX and risk free rates is recorded in the output, but the exact observation timestamp of every automatically retrieved market input is not retained as a separate audit field. Full historical reproducibility of all external inputs is therefore limited.

The fallback market data snapshot improves operational robustness but can become stale. Fallback risk free rates are intentionally simplified and may represent flat market or policy rate proxies rather than calibrated zero coupon yield curves.

The methodology used to obtain risk free rate proxies also differs across currencies. This is sufficient for the demonstration framework but is less internally consistent than constructing comparable market curves using a common methodology.

FX rates are used only to translate local currency exposures into the selected base currency at

the start of the calculation. FX movements over the one year risk horizon are not simulated.

Consequently, the model measures credit risk on normalized exposures but does not measure the additional market risk created by foreign currency positions. Credit and FX risk are also not modeled jointly.

7.9 Monte Carlo Simulation and Tail Risk Measures

VaR and Expected Shortfall are estimated from a finite Monte Carlo sample and are therefore subject to simulation error.

This effect becomes increasingly important at high confidence levels. Under the baseline configuration of 100,000 simulations and a 99.5% confidence level, only 500 scenarios contribute to the Expected Shortfall tail. At a 99.9% confidence level, only 100 observations remain.

The Monte Carlo stability analysis demonstrates that estimates can vary across random seeds, particularly at lower simulation counts. Increasing the number of simulations improves overall stability, but it does not remove structural model uncertainty.

The stability experiment uses three random seeds for each simulation size. This is sufficient to illustrate sampling variability, but it is not a formal statistical convergence study. A more extensive analysis could use many independent repetitions and construct confidence intervals for VaR and Expected Shortfall.

VaR has the additional limitation that it describes a quantile rather than the severity of losses beyond that quantile. The concentration analysis illustrates this clearly: a portfolio can have a relatively low VaR while still producing severe losses further into the tail.

Expected Shortfall provides additional information by averaging extreme losses, but it is correspondingly more dependent on the small number of scenarios in the far tail and can therefore be numerically noisy.

The reported risk measures are conditional on the modeled scenario distribution. They should not be interpreted as substitutes for separate stress testing, reverse stress testing or scenario analysis involving specific historical or hypothetical crises.

7.10 Counterparty Risk Contributions

The framework reports standalone, incremental, marginal and diversification measures for individual counterparties. These quantities are useful for understanding portfolio risk drivers, but their interpretation requires care.

Incremental risk is calculated using a leave one out approach:

$$\rho(L) - \rho(L - L_i). \quad (43)$$

The calculation therefore measures the effect of removing the entire counterparty from the existing portfolio rather than the effect of an infinitesimally small change in exposure.

The reported marginal measure is obtained by dividing this leave one out incremental risk by the counterparty exposure. It is therefore an exposure normalized incremental measure and not the mathematical derivative of the risk measure with respect to exposure.

In particular, it is not an Euler allocation of portfolio VaR or Expected Shortfall. The individual incremental or marginal measures do not generally sum to total portfolio risk.

The measures also depend on the current portfolio composition. Removing a counterparty changes its interaction with every remaining counterparty, so the result reflects both the counterparty’s standalone risk and its diversification or concentration effects.

VaR based contributions can additionally be sensitive to small changes around the portfolio quantile because the underlying loss distribution is discrete and simulated. Expected Shortfall contributions are generally more informative about the extreme tail but remain subject to Monte Carlo sampling error.

The contribution measures should therefore be interpreted as diagnostic portfolio risk indicators rather than as a formal additive economic capital allocation.

7.11 Validation Scope

The validation framework challenges several important assumptions: dependence model, Student t-copula degrees of freedom, LGD, concentration, factor structure, simulation size, random seed and confidence level.

These tests provide evidence that the implementation behaves coherently when key assumptions are changed. They also identify the parameters to which the resulting risk measures are particularly sensitive.

However, sensitivity analysis is not equivalent to empirical model validation. The project does not contain a sufficiently large historical portfolio of realized rating migrations and defaults against which predicted probabilities and portfolio loss distributions could be backtested.

The rating transition matrix, rating PD mapping and credit spreads are synthetic. Consequently, there is no meaningful statistical backtest of their calibration against historical agency or market data within this demonstration.

Similarly, the factor model has not been validated through a dedicated out of sample forecasting exercise, and the copula parameters have not been calibrated and tested against an independent credit event dataset.

The concentration sensitivity is also not a pure test of concentration in isolation. Changing the Apple exposure while rescaling the remaining portfolio simultaneously changes the portfolio’s credit quality composition. The experiment is informative, but its effects therefore combine concentration and credit mix.

Automated unit tests and experiment integrity checks provide strong implementation controls, but successful software tests cannot establish that the economic assumptions of the model are correct. They verify that the implemented methodology behaves as designed.

7.12 Implications for Model Use

The framework is most appropriately used as a transparent quantitative laboratory for studying how structural credit risk, rating migration, systematic dependence, concentration and tail risk interact.

Its main strength is not that it eliminates model uncertainty, but that the individual assumptions are explicit, configurable and testable. The validation results show that several of these assumptions, particularly dependence, tail dependence, LGD and concentration, can materially affect the resulting portfolio risk measures.

A production extension would require substantially stronger empirical calibration and data

governance. Important enhancements would include empirical rating transition and spread data, richer structural default modeling, stochastic and counterparty specific LGD, dynamic exposure at default, full risk free and credit spread term structures, more comprehensive point in time market data handling, calibrated dependence parameters, additional stress scenarios and historical backtesting.

The limitations therefore do not invalidate the framework’s purpose. Rather, they define the boundary between a transparent demonstration model and a production credit risk system and identify the areas in which additional modeling sophistication would provide the greatest benefit.

8 Conclusion

This project developed a transparent Python framework for portfolio credit risk analysis that combines structural default modeling, rating based credit risk, rating migration, systematic dependence and Monte Carlo tail risk measurement within a single workflow.

At the counterparty level, probabilities of default can either be supplied through a rating based approach or estimated using the structural Merton model. This allows the same portfolio to contain counterparties with externally specified credit quality and counterparties for which default risk is derived from equity and balance sheet information. Structural probabilities of default are subsequently mapped into equivalent rating categories so that all counterparties can enter a common rating migration framework.

The portfolio model extends default only analysis by allowing counterparties to migrate between rating states over a one year horizon. Non-default migrations affect portfolio value through rating dependent credit spread revaluation, while default losses are determined using exposure and LGD. The resulting loss distribution therefore captures both default events and mark-to-market effects caused by changes in credit quality.

Dependence between counterparties is introduced through global, regional and sector factors together with idiosyncratic risk. The framework supports independent outcomes, Gaussian dependence and Student t-copula dependence. This makes it possible to examine not only individual counterparty risk but also the effect of systematic credit conditions and joint tail events on the portfolio.

For the baseline demonstration portfolio, total exposure is approximately USD 899.59 million. Default expected loss is approximately USD 1.17 million, while the mean simulated portfolio loss including rating migration and spread revaluation is approximately USD 5.14 million. At a confidence level of 99.5%, portfolio VaR is approximately USD 78.31 million and Expected Shortfall is approximately USD 141.13 million.

The difference between default expected loss and the complete simulated loss measures is economically important. It demonstrates that portfolio credit risk is not determined by default probabilities alone. Rating deterioration, spread revaluation, exposure concentration and systematic dependence can materially affect both expected losses and the shape of the portfolio tail.

This distinction is also visible at the individual counterparty level. A very small structural Merton probability of default does not necessarily imply a negligible portfolio risk contribution. Once a structural PD is translated into an equivalent rating, the counterparty remains exposed to migration and spread revaluation. The framework therefore distinguishes between structural default risk and the broader economic effects associated with deterioration in credit quality.

The validation analysis provides additional insight into the main drivers of portfolio risk. De-

pendence assumptions have a particularly strong influence on the tail of the loss distribution. Moving from independent credit outcomes to Gaussian dependence and then to the Student t-copula materially increases both VaR and Expected Shortfall while leaving the mean loss comparatively stable. Within the Student t-copula specification, lower degrees of freedom generate substantially larger tail losses.

LGD sensitivity behaves in the expected direction, with higher LGD assumptions increasing default expected loss, VaR and Expected Shortfall. The factor structure also affects the resulting tail distribution, demonstrating that the choice and calibration of systematic risk drivers represents a relevant source of model risk.

The concentration analysis highlights an especially important distinction between VaR and Expected Shortfall. Increasing the portfolio weight of a relatively high quality counterparty can reduce default expected loss and the VaR quantile while simultaneously increasing losses in the most extreme scenarios. In the most concentrated experiment, the 99.5% VaR falls to approximately USD 49.03 million while Expected Shortfall rises to approximately USD 182.26 million. This illustrates why a single quantile is not sufficient to describe portfolio concentration and extreme loss severity.

The Monte Carlo stability analysis further shows that numerical uncertainty becomes important when tail risk is estimated from a limited number of scenarios. Estimates from 10,000 simulations vary materially across random seeds, whereas the 250,000 scenario experiments are considerably more stable. The baseline choice of 100,000 simulations therefore represents a practical compromise for the demonstration model, while the validation framework makes the remaining simulation uncertainty visible.

An important objective of the project was also to treat implementation quality as part of quantitative model development. The framework separates reusable model logic from examples, reporting and validation code. Portfolio assumptions are controlled through an external Excel steering file, model inputs are validated before execution, random seeds can be fixed for reproducibility, and automated tests and experiment integrity checks are used to identify implementation errors. The final validation calculation also reproduces the baseline 99.5% VaR and Expected Shortfall exactly from the stored scenario losses.

These controls improve transparency and reproducibility, but they do not eliminate economic model risk. The framework remains a demonstration model built using synthetic credit inputs and several simplifying assumptions. Structural PD estimation, rating migration, credit spread revaluation, LGD, systematic dependence and market data treatment all contain approximations that would require substantially stronger empirical calibration and governance in a production environment.

The main contribution of the project is therefore not a claim that a single model provides a definitive measure of credit risk. Instead, it demonstrates how several important components of modern credit portfolio analysis can be connected within a coherent and testable quantitative framework. Structural default risk, rating migration, dependence, concentration and tail risk can be examined jointly while the assumptions responsible for the resulting risk measures remain visible and configurable.

The framework also provides a natural basis for further development. Potential extensions include empirical rating transition and credit spread data, richer structural default models, stochastic and counterparty specific LGD, dynamic exposure at default, full risk free and credit spread term structures, stronger point in time market data handling, empirical dependence calibration, alternative copula structures, stress testing and historical backtesting. These extensions would move the framework from a transparent demonstration model toward a more comprehensive credit portfolio risk system.

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